التكنولوجيا التطبيقية
APPLIED TECHNOLOGY
وزارة التربية والتعليم والتعليم الفني

جمهورية مصر العربية
وزارة التربية والتعليم والتعليم الفني
وحدة تشغيل وإدارة مدارس التكنولوجيا التطبيقية

Student guide

Established Chemistry

Class:First stage of Secondary School

Preparation

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Reviewer

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Unit one: The Trend of Elements Properties in the Modern Periodic Table

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Unit one: The Trend of Elements Properties in the Modern Periodic Tabl

Dear student by the end of the following unit you should have the following abilities and knowlages:

The abilities: To be able to:

1-Distinguish between the type of elements(metals , nonmetals and metalloids)

2- Distinguish between oxides of metals, nonmetals and metalloids

3- Compare between different properties for the elements for the elements (atomic radius , ionization potential ,.....etc.)

4- Calculate the oxidation number of the elements in some compounds

Knowledge:To learn about :

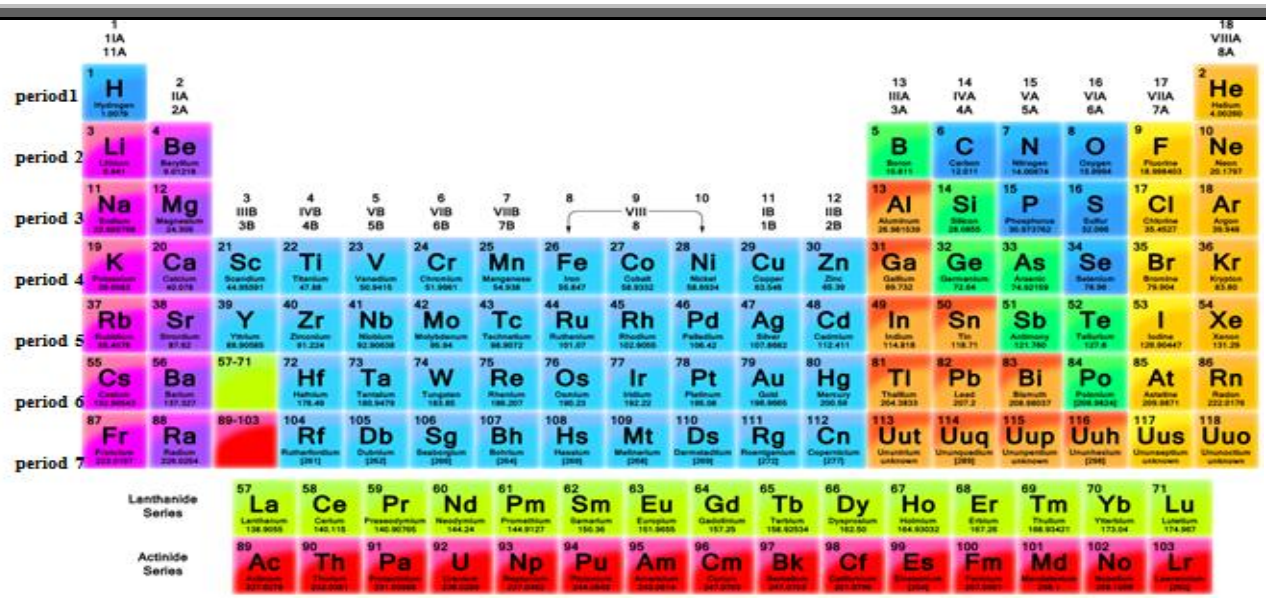
1- Description of the periodic table

2- The diameter of the elements and its trend in the modern table

3- Metallic, nonmetallic and metalloid properties of the elements in the table

4- Oxidation numbers of the elements

lesson one
Modern Periodic Table



The well known elements are arranged in a table called modern periodic table.

- The elements are arranged according to the increasing of their atomic number.(number of protons inside the nucleus of the atom)

1-In eighteen Vertical columns each is called group and seven horizontal periods each is called period.

Each element has one electron more than the element before it.

2- four blocks s , p , d and f blocks

Description of long periodic table

S– block elements

- It consists of two groups 1A and 2A

P – block elements

- It consists of six groups in the right side of the table
- it includes the groups 3 A , 4A , 5A , 6 A , 7 A and zero

G.R : Both $_{17}Cl$ and $_{36}Kr$ are from P block

d – block elements

- It consists of (10) groups it is found in the middle of the table
- it includes the groups 3 B,4B,5B,6B,7B , 8 , 1B and 2B
- it includes three series:

The first transition series



The second main transition series:



The third main transition series:



F–block elements

It consists of (14) elements,

They are found below of the table

These elements are separated from the table to be not so wide

It includes lanthanides and actinides

Lanthanides :



↳ So they have similar properties and difficult to be separated from each other's so they are called rare earth elements

Actinides :



They are radioactive elements and their nuclei are unstable and produces radiations such as alpha, Beta and gamma



Types of the elements

1-Noble gases:

- *They are the elements of the last column .*

↳ All of their energy levels are completely filled with electrons.

↳ They are unreactive elements under normal conditions.

↳ They are mono atomic molecules

2-The representative elements:

They are the elements of main group (S,P- blocks) except inert gases

↳ All of their energy levels are completely filled with electrons except for the **external energy** level.

↳ Tend to reach their electronic structure to the nearest noble gas

3-The transition elements:

↳ They occupy the middle of the table .

They are the elements of the d – block

4-The inner transition elements:

- *They are the elements of the f – block*

- **The long form periodic table**: the elements were arranged in an ascending order according to their atomic numbers

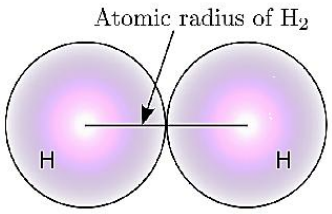
The periodic table is divided into four main blocks

1- S – block 2- P – block 3- d – block 4- f – block

Trends and periodicity of properties in the periodic table

The atomic radius:

- We can't determine the atomic radius because electron has a wave motion so it is impossible to determine exactly the location of an electron around the nucleus.



The atomic radius: It is half the distance between centers of two similar atoms in a diatomic molecule.

The bond length: It is the distance between nuclei of two bonded atoms.

✓The methods used to measure the bond length such as:

- 1- X – ray. 2- Electron diffraction.

The crystals of ionic compounds are construction of anions and Cations .The distance between the centers of cation and anion can be measured .This distance depends on the number of electron lost or gained .

Problem The bond length in the chloride molecule Cl – Cl is 1.98 Å and the length between carbon and chloride atoms C – Cl is 1.76 Å Calculate the atomic radius of carbon .

The atomic radius of chlorine = $\frac{1.98}{2}$ = 0.99 Å

The atomic radius of carbon = 1.76 – 0.99 = 0.77 Å

Problem The bond length in the molecule of NH_3 is 1.0Å and the bond length in the molecule of H_2 is 0.6Å Calculate the bond length in (N_2)?

The atomic radius of hydrogen = $\frac{0.6}{2}$ = 0.3 Å

The atomic radius of nitrogen = 1 – 0.3 = 0.7 Å

The bond length of the N_2 = 0.7 × 2 = 1.4 Å

The graduation of atomic radius :

- The radius is inversely proportion to the attraction forces between the nucleus and the outer electrons.
- The effective charge of nucleus" Z_{eef}": it is the actual charge affecting the electron is always less than the nuclear charge because the screen effect of the completed energy levels *decreases the nuclear attraction force on the outer electrons*

G.R.: *The effective charge of nucleus always less than nucleus charge*

In period: Atomic radius decrease with increasing atomic number

Due to: *Increasing of effective nuclear charge from left to right*

The largest group 1A as sodium and smallest halogens group 7A as chlorine

In group:

Atomic radius increases with increasing of atomic number from up to down due to

1-Increasing of the energy levels

2-The completed inner levels screen the nucleus effect on the outer electrons

3-Increasing the repulsion between electrons of the complete levels

the relation between the radius of ion and the radius of its atom:

The radius of cation (+ve ion) is smaller than that of its atom

The anion's radius is bigger than that of its atom

How can you explain the change in the following radii :

Atom or ion	Na	Na^+	Fe	Fe^{2+}	Fe^{3+}	Cl	Cl^-
Radius A^0	**1.57**	**0.95**	**1.17**	**0.75**	**0.60**	**0.99**	**1.81**

Lesson -2

Ionization potential

► ***It is the amount of energy required to remove the most loosely electron completely from single and gaseous atom***

✓The electron is attracted to the atom by attraction force

To remove the electron completely it needs to gain an amount of energy. This energy called ionization potential or energy

Atom M ⟶ cation M^+ + e^- ΔH = + ve KJ/Mole

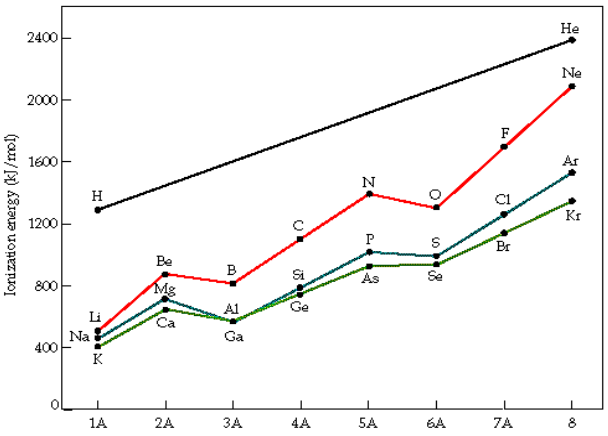
✓The atom losses electrons and converted into positive ions. It has a positive value (ΔH = + ve KJ/Mole) .

The first ionization energy

○ fdcIt is the energy required to remove one electron from neutral atom to form a cation (+ ve) with one positive charge

$M \longrightarrow M^+ + e^-$

ΔH = + ve KJ/Mole



In the period

Ionization energy increases with increasing the atomic number

Because the decrease the atomic radius and increase the attractive force of the nucleus on the valence electrons

So the electrons need high energy to be removed (or separated) from the atom.

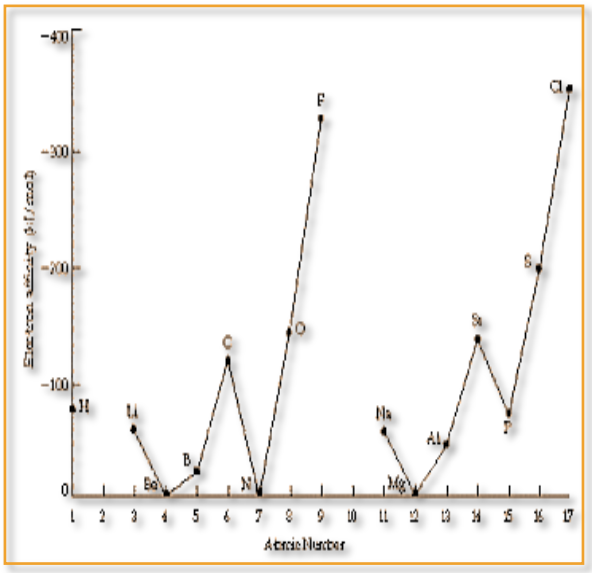
In the group :

Ionization energy decreases with increasing the atomic number

Due to increasing in the atomic size and decreasing the attraction force of the nucleus to the valence electrons therefore the electrons need a smaller value of energy to separate from the atom.

The electron affinity: It is the amount of energy released when an extra electron is added to a neutral gaseous atom to form an anion (-ve ion)

$X + e^- \longrightarrow X^-$ + energy ΔH = - ve KJ/Mole



G.R : In the horizontal periods electron affinity increases with the increasing of atomic number

Due to the atomic **radius** (size) gradual **decrease** so it **becomes easier** for the **nucleus** to **attract** the extra electron.

The electron affinity decreases in group:

Due to the increase of the atomic volume with increase atomic number and this leads to the decrease of the attraction force of the nucleus on the valence electrons.

G.R :Noble gases have not (too small) electron affinity?

Because they have completely filled levels with electrons which gives the atoms great stability.

G.R :Electron affinity of Fluorine (F_9) is less than that of chlorine (Cl_{17}) ?

- Because the atomic radius (size) of fluorine atom is smaller than that of chlorine atom and the extra electron is affected by a greater repulsion force than that in chlorine atom .
- **Electron affinity increases, if the gained electron makes the outer sublevel of the atom filled or half filled**
- Electron affinity decreases, if the outer sublevel is completely filled or half filled for the atom

Electronegativity:

It is the tendency of an atom to attract the electrons of chemical bond to itself.

G.R :Electro negativity increases in the period?

Because the positive nuclear charge gradually increases with the increase of the atomic number leads to decrease the atomic radius and increase the attractive force of the nucleus

G .R :The electronegativity decreases in group?

- Due to the increase of the atomic volume with increase atomic number and this leads to decrease of the attraction force of the nucleus to the extra electron

Compare between ionization energy , electron affinity and electro negativity

Ionization energy	Electron affinity	Electro negativity
It is the amount of energy needed to remove the most loosely electron in a neutral single gaseous atom	It is the amount of energy released when an extra electron is added to a neutral single atom	It is tendency of an atom to attract the electrons of chemical bond to itself
It refers to the atom in its single state	It refers to the atom in its single state	It refers to the atoms which linked together in the molecule
It is inversely proportional to the atomic radius	It is inversely proportional to the atomic radius	It is inversely proportional to the atomic radius

Lesson-3
Metals and Non-Metals

There are three types of elements in the periodic table:
metals - nonmetals and metalloids.

Metals:

- Their Valence shell has less than half capacity of electrons (1 or 2 or 3).
- They one called electropositive elements.
- They have Relatively large atomic radius therefore the ionization energy and electron affinity and electro negativity have smell values.
- They have a good conductivity of electricity.

Non metals

- Valence shell has more than half its capacity of electrons(5, 6 or 7).
- They are called electronegative elements.
- They have small atomic radius ionization energy and electro negativity and electron affinity have high value.
- They have a bad conductivity of electricity.

G.R:Fluorine is the **strongest** non-metal while **cesium** is the **strongest** metals?

Because fluorine has smallest radius while cesium has the biggest radius.

G.R: Metals are considered as electropositive elements ?

Because metals lose electrons to form positive ions.

G.R: Non-metals are considered as electronegative elements?

Because non- metals gains electrons to from negative ions.

G.R: Metal are good conductors of electricity?

- Because they have free valence electrons that can transfer easily from one position to another in the metal structure conducting the electricity.

G.R: Non-metals are bad conductors of electricity?

- Because their valence electrons are strongly bounded to the nucleus due to the small atomic size therefore it is difficult for the valence electrons to be transfer.

G.R: Metals have small values for ionization energy and electron affinity?

- Because they have large atomic radius.

G.R: The strongest metal lies at the bottom on the left and side of the periodic table ?

- Because of the increasing in the atomic number the atomic radius increases gradually so the attractive force of the nucleus to the valence electrons decreases therefore it is very easy for the atom to lose the valence electrons so the metallic property increases.

Metalloids

- They are elements whose have the appearance of metal and most of nonmetals properties . Sometimes they act as metal (when they lose electrons).

They act as semiconductors (Boron – silicon) which are used in transistors and known as semiconductors electronic instruments.

G.R: Fluorine is considered as the strongest non–metal?

Because the atomic size of fluorine is very small therefore the attractive force of the nucleus to the electrons will be increased therefore it is very easy for the atom to gain a new electrons .

The oxide is formed when the element combines with oxygen

Classification of oxides: acidic – basic and amphoteric

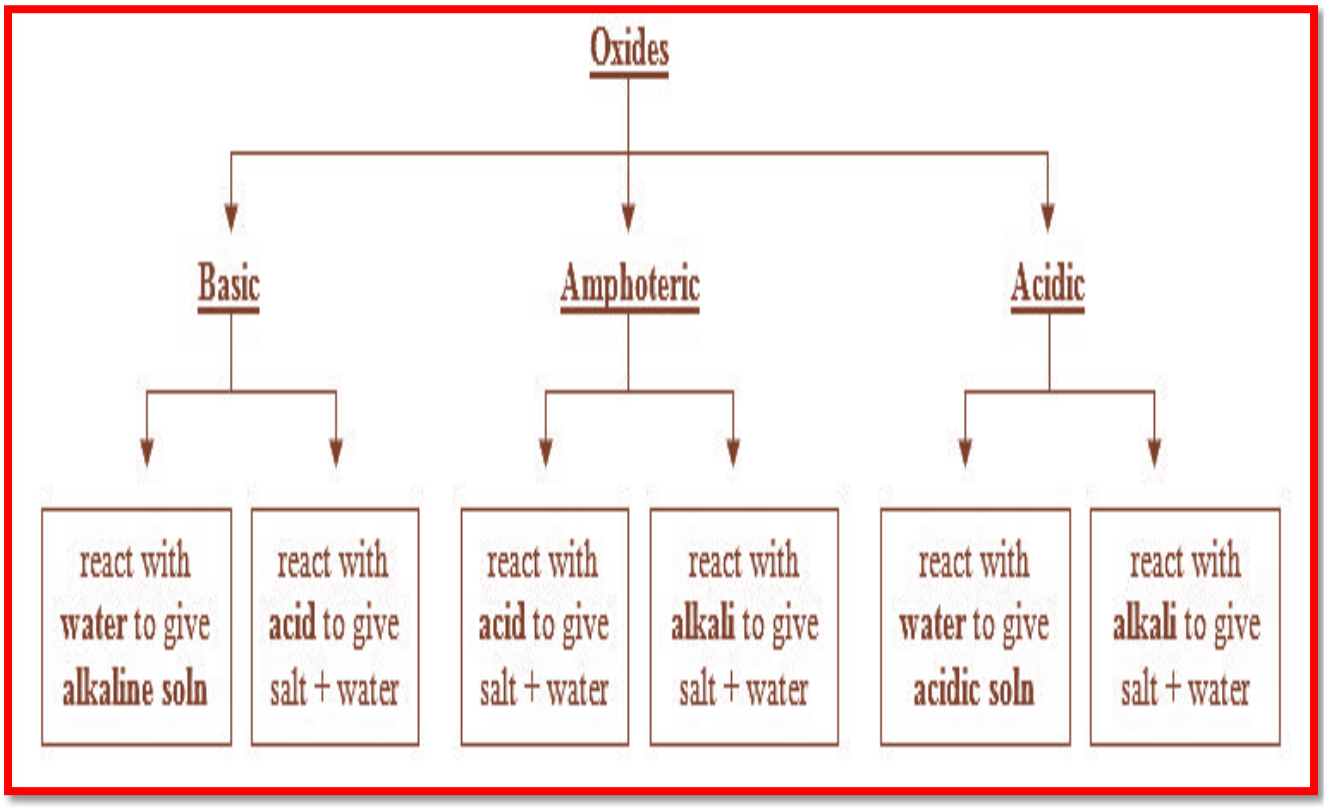
1. Acidic oxides	2. Basic oxides
They are non – metal oxides such as CO_2,SO_2,SO_3 , P_2O_5	They are metal oxides such as Na_2O,K_2O,MnO,CaO,BaO
They dissolve in water to form acids $CO_2 + H_2O \rightarrow H_2CO_3$ $SO_3 + H_2O \rightarrow H_2SO_4$ $P_2O_5 + H_2O \rightarrow 2H_3PO_4$	Some basic oxides dissolve in water to form alkalis $Na_2O + H_2O \rightarrow 2NaOH$ $K_2O + H_2O \rightarrow 2KOH$ $MgO + H_2O \rightarrow Mg(OH)_2$
They react with alkalis to form salt and water $CO_2 + 2NaOH \rightarrow Na_2CO_3 + H_2O$	They react with acids to form salt and water $Na_2O + 2HCl \rightarrow H_2O + 2NaCl$
They do not reaction with acids	They do not react with alkalis

3- Amphoteric oxides :

They react with bases and acids to form salt and water in both cases

They have both acidic and basic properties such as Al_2O_3 , ZnO , Sb_2O_3 , SnO .

- $ZnO_{(S)} + H_2SO_4 \rightarrow ZnSO_{4(aq)} + H_2O_{(l)}$
- $ZnO_{(S)} + 2NaOH_{(aq)} \rightarrow NaZnO_{2(aq)} + H_2O_{(l)}$



Mr. Nabil's Notes in chemistry Graduation of acidic or basic property in periodic table:

In the period

- Acidic property of oxides increases with increasing the atomic number.

In the group :

- Acidic property of oxides decreases with increasing the atomic number.

Oxygenated acids (Oxyacid) are :

- acids that contain hydrogen, oxygen and a third element usually a non-metal.

- It can take the following symbols: - $[MO_n(OH)\ m]$ **Where,**

- (M) Is the atom of the element. - (n)Is the number of oxygen atoms.
- (m) Is the number of hydroxyl groups .
- The **strength** of oxyacid depends on the **number of oxygen atoms** which isn't linked with hydrogen atoms when this number of free Oxygen atoms increase,the strength of the acid.

Acid	Free O atoms	Acidic Property
Orthosiliconic H_4SiO_4	$Si(OH)_4$ = zero	Weak acid
Phosphoric H_3PO_4	$PO(OH)_3$ =1	Moderate acid
Sulphuric H_2SO_4	$SO_2(OH)_2$= 2	Strong acid
Perechloric $HClO_4$	$ClO_3(OH)$= 3	Very strong acid

How can you obtain CuO from mixture of ($CuO+ Al_2O_3$)?

G.R:KOH is stronger base than of NaOH?

- Because the atomic radius of K is greater than that of Na so the bond strength between K and hydroxide group (OH^-) decreases therefore it is easy to separate hydroxide ions
- The strength of acid increases with increasing of the ability of acid to lose the proton H^+

G.R:HF is weak acid ?

- Because the atomic size of fluorine atom is very small so it has great ability to attract the electrons of bond to itself therefore it is difficult to separate hydrogen ions.

G.R:HI is strong acid ?

- Because the atomic (size) radius of iodine atom is large to it has a weak ability to attract the electron bond to itself therefore it is easy to separate hydrogen (H^+) ions to form strong acid .

Lesson -4

Oxidation Numbers

It is a number that refers to the electric charge (+ve or -ve) that atom would have in the compound .

- In ionic compounds = the valence of the elements with a sign which is the charge of the ion
- +ve means the atom lost electrons and –ve means the atom gained electrons
- In covalent compounds = the number of electrons shared by the elements with a sign which depends on the electronegativity of the element or displacement of electrons.
- +ve sign to the atom of lower electro negativity and –ve sign to the atom of higher in electro negativity

Rules for assigning oxidation numbers

Oxygen	Has (-2) in most of its compounds except in 1. peroxides (H_2O_2) and Na_2O_2 is (1-) 2. The oxidation number of oxygen = (2+) in OF_2 Because the electro negativity of fluorine is higher than the electron negativity of oxygen.
Hydrogen	Always (1+) Except in hydrides =(1-) such as NaH , KH , CaH_2
Group 1A	↳ always (1+) such as Na , K and Li
Group 2A	↳ always (2+) such as Mg , Ca
Group 3A	↳ always (3+) such as Al
Group 7A	✓always (-1) such as Cl , Br , I and F

	except with oxygen it will be (1+) except fluorine always -1
compound	= zero Such as NaCl, HCl, $CuSO_4$
element	= zero. e.g. Fe , Cl , Na , O_2 , S , P_4 .
atomic group	= the number of charge on the group (ion) such as OH^- = -1 , SO_4^{--}= -2 , CO_3^{--} =-2 , NH_4^+ =+1, PO_4^{---} =-3

The oxidation process: it is the process of losing electrons due to increase the oxidation number for the element.

The Reduction process: it is the process of gaining electrons due to decrease the oxidation number for the element.

Oxidation	Reduction
Zero ⟶ (+) value (+) small value⟶ (+) big value (-) value ⟶ zero (-) big value ⟶(-) small value	(+) value ⟶ zero (+) big value⟶ (+) small value zero ⟶ (-) value (-) small value ⟶ (-) big value

Explain the type of change (oxidationor reduction) that of chromium and iron in this reaction.

$K_2Cr_2O_7 + 6FeCl_2 + 14HCl \rightarrow 2KCl + CrCl_3 + 6FeCl_3 + 7\ H_2O$

$K_2Cr_2O_7 \longrightarrow CrCl_3$	$FeCl_2 \longrightarrow FeCl_3$

$Cr^{+6} \xrightarrow{\text{oxidation}} Cr^{+3}$	$Fe^{+2} \xrightarrow{\text{reduction}} Fe^{+3}$
Cr^{+6} is reducing agent	Fe^{+2} is reducing agent

Gradation of oxidation number in periodic table:

- The oxidation number of element (metal) of 1st and 3rd groups in their compounds agrees with group number (It takes positive oxidation number to which they belong)
- Most elements in the middle of the table have variable oxidation number in different compounds.
- The oxidation number of the elements in groups from IV to VII ,

Equals(the group number -8) = - value

Because the elements gain electrons to complete a full shell

Lesson-5 Evaluation

EVALUATION

QUESTION 1:

Choose the correct answer :

1- Period six in the periodic table contains ..... types of elements

a. six b. three c. four d. five

2- Nonmetals are characterized by .....

a. large ionization energy

b. electropositive elements

c. small electron affinity

d. large atomic radius

3- Electronegativity increases across the periods by .....

a. increasing atomic radius

b. decreasing the atomic number

c. decreasing atomic radius

d. (a and b) correct.

6- One of the following reactions does not represent an oxidation reduction reaction , it is no .....

a. $2P + 5HC\ell O + 3H_2O \longrightarrow 2H_3PO_4 + 5HC\ell$

b. $Zn + 2HC\ell \longrightarrow ZnC\ell_2 + H_2$

c. $Mg + CuSO_4 \longrightarrow MgSO_4 + Cu$

d. $NaOH + HNO_3 \longrightarrow NaNO_3 + H_2O$

7- One of the following reaction represents an oxidation reduction reaction , it is no .....

a. $CuO + H_2SO_4 \longrightarrow CuSO_4 + H_2O$

B. $CaCO_3 + 2HC\ell \longrightarrow CaC\ell_2 + H_2O + CO_2$

c. $Cr_2O^{2-}_7 + 3H_2S + 8H^+ \longrightarrow 2Cr^{3+} + 3S + 7H_2O$

d. $NaC\ell + AgNO_3 \longrightarrow AgC\ell + NaNO_3$

QUESTION 3:

What is meant by each of the following:

1- Atomic number
2- Reduction
3- Representative element
4- Nobel element
5- Transition element
6- Inner transition element
7- Atomic radius
8- ionization potential
9- Electron affinity
10- Electronegativity
11- Metals
12- Non metals
13- Metalloids
14- Acidic oxide
15- Basic oxide
16- Amphoteric oxide
17- Oxidation number
18- Oxidation

QUESTION 4:

Describe the graduation of the following properties in the periodic table.

a. atomic radius
b. ionization energy
c. electron affinity
d. electronegativity
e. metallic and nonmetallic property
f. acidic and basic property of the elements of the third period

QUESTION 5:

If you know that:

The bond length in hydrogen Fluoride molecules equal (0.94 A°)

The bond length of hydrogen molecule is (0.6 A°), calculate the bond length in fluorine molecule.

QUESTION 6: Explain the differences between:

electron affinity and electronegativity

metals and non metals

acidic, basic and amphoteric oxide

QUESTION 7:

What is the scientific term for each of the following :

a. half the distance between the centers of two similar atoms in a diatomic molecule.

b. the amount of energy needed to remove the most loosely bound electron completely from an isolated gaseous atom.

c. the amount of energy released when an extra electron is added to neutral gaseous atom.

d. the tendency of an atom to attract the electrons of the chemical bond to itself.

e. the group of elements whose valence shell has less than half its capacity of electrons.

f. the group of elements whose valence shell has more than half its capacity of electrons.

g. A number that refers to the electric charge that the atom would have in the compound.

h. the process of losing electrons associated with increasing the positive charge

i. the process of gaining electrons and associated with a decrease in the positive charge.

QUESTION 8:

Calculate the oxidation number of the mentioned element in following compounds :

a. oxygen in : O_2 - O_3 - Li_2O - Na_2O_2 - KO_2 - OF_2.

b. chlorine in : Na $C\ell$ - Na $C\ell O_4$ - Na $C\ell O_3$ - Na $C\ell O_2$ - Na $C\ell O$.

c. nitrogen in :NH_3 - N_2 - N_2O - NO - HNO_2 - NO_2 - HNO_3.

d. sulphur in : $Na_2S_2O_3$ - K_2S - SO_2 - $NaHSO_3$ - H_2SO_4.

e. manganese in : MnO_2 - $KMnO_4$ - $MnC\ell_2$ - $NaMnO_4$.

Unit Two: The Solutions

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Unit two: The Solutions

Dear student by the end of the following unit you should have the following abilities and knowlages:

The abilities: To be able to:

Distinguish between the different types of solutions

Differentiate between electrolytic and non-electrolytic solutions

Distinguish between the different types of solutions by using indicators

Compare between strong and weak acids or bases

.Knowledge:To learn about :

1- The collective properties of the solutions

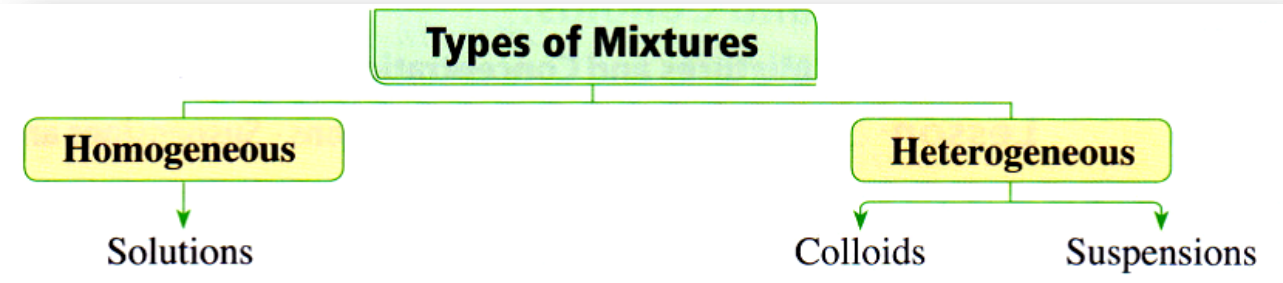
2- The properties of different solutions (true , colloidal and suspended)

3 – Classifications of acids and bases

Lesson -1
The Solutions

Solutions

✓ The mixtures are classified according to homogeneity into two types.



Homogeneous	Nonhomogeneous	
solution	**suspensions**	**Colloid**
Their components cannot be distinguished by the naked eye or the electron microscope. **Examples** (Table salt- Cane - sugar Cobalt (II) chloride) solutions in water.	Their components can be distinguished by the electronic microscope only. **Examples** 1. Aerosols. 2. Hair gel. 3. Mayonnaise. 4. Blood.- Milk.	Their components can be distinguished by the naked eye. **Examples** 1. Table salt in kerosene. 2. sugar in kerosene. 3. Cobalt (II) chloride in kerosene. 4. Oil in water.

Homogeneous mixtures

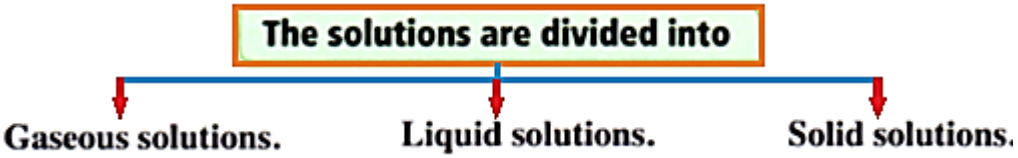
If you add a small quantity of a substance (as sugar) to a large amount of another substance (as water) and disappears in it.

Therefore:• Sugar is a solute. • Water is a solvent. • The produced mixture is a solution.

solvent:it exists in greater amount in solutions

Solute: it exists in smaller amounts in solution.

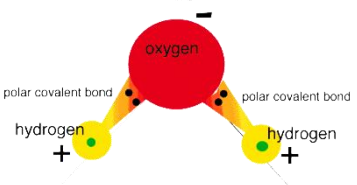
Solutions are classified according to the physical state of solvent:



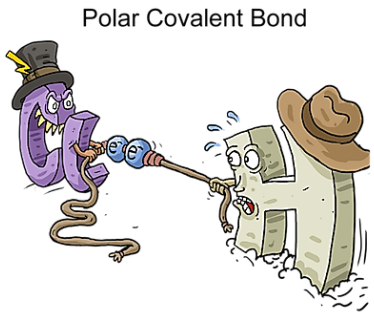
Solution type	Solvent	Solute	examples
Gaseous	Gas	Gas	Air – natural gas
Liquid	Liquid	Gas	Dissolved oxygen in water-Fizzy drinks
		Liquid	Alcohol in water- ethylene glycol in water
		solid	salt of sugar in water
solid	solid	Gas	Hydrogen in palladium and platinum
		Liquid	silver amalgam Ag(s) / Hg(l)
		solid	alloys (Nickel chrome alloy)

Some important concepts

Electronegativity: The ability of atom to attract electrons of the chemical bonds in the molecule to itself.



Polar bond: A type of covalent bond between two atoms of different electronegativity . The atom of high electronegativit carries a partial negative charge (δ^-) and that of less electronegativity carries a partial positive charge (δ^+)



- **Water** is the most famous **polar solvent** and most of solutions are prepared in the presence of water.
- The water molecule is composed of **oxygen** atom linked to **two hydrogen** atoms by **two single** bonds and the angle between the two bonds equals 104.5°
- **The polarity** of water molecule is due to :

✓ The ability of the oxygen atom to attract electrons of bonds is greater than that of the hydrogen atom.

✓ A partial negative charge is formed on the oxygen atom and a partial positive charge on the hydrogen atoms so the water molecule is polar .

Solutions as Electrolytes and Non electrolytes

➢ **Electrolytes:** Substances whose solutions or molten liquids conduct electric current via the movement of free ions

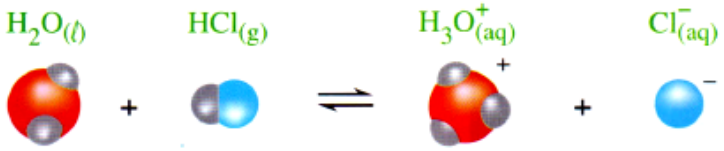
➢ Strength of Electrolytes is measure by the number of free ions

➢ Electrolytes are divided into:-

Strong Electrolytes	Weak Electrolytes
The solute is completely dissociated in water into free ions,and it is good conductors of electricity	The solute is partially dissociated in water (small no. of its molecules dissociate in water) and it is poor conductors of electricity
Examples 1-Ionic compounds: - sodium chloride (NaCl) sodium hydroxide (NaOH)	Examples - 1. Acetic acid CH_3COOH (vinegar) 2. - Water H_2O 3. -Ammonium hydroxide solution NH_4OH
2-Polar covalent compounds: such as (HCl) in water	

- When hydrochloric acid dissolves in water, positive hydrogen ions (H^+)are produced and bond with water molecules forming hydronium ions **(H_3O^+)**

$$HCl_{(g)} + H_2O_{(l)} \rightarrow H_3O_{(aq)} + Cl^-_{(aq)}$$



- The H^+ ions doesn't exist freely in water because they combine with water forming hydronium ions**(H_3O^+)**

G . R : Sodium chloride solution is strong electrolyte while Acetic acid solution is weak electrolyte

THE ANSWER Sodium chloride molecules are completely dissociate in water into ions while acetic acid molecules are partially dissociate into ions in water

$$NaCl_{(s)} \xrightarrow{H_2O} Na^+_{(aq)} + Cl^-_{(aq)}$$

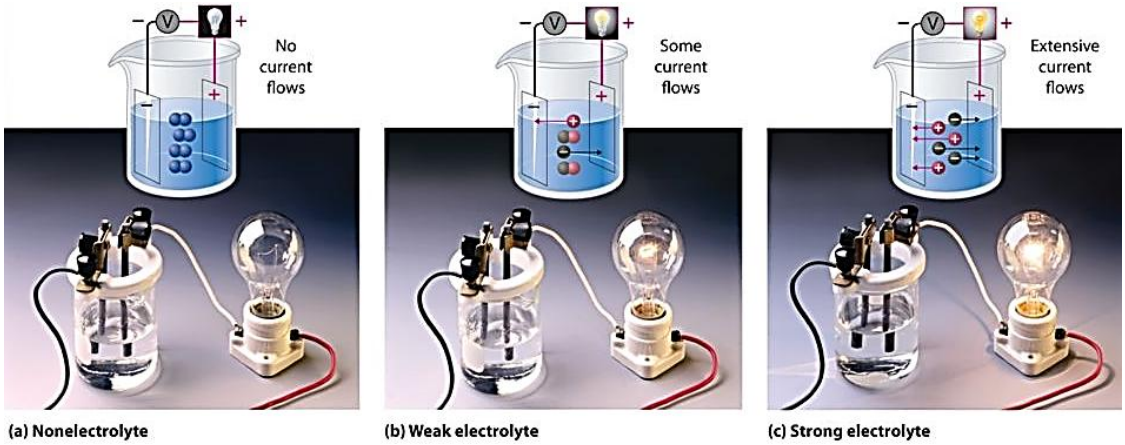
$$CH_3COOH_{(s)} \rightleftharpoons^{H_2O} CH_3COO^-_{(aq)} + H^+_{(aq)}$$

Non electrolytes: Substances whose solutions or molten liquids don't contain free ions so they don't conduct electricity

- **Examples**: Sugar or Ethyl alcohol (ethanol) in water

G . R : Sugar solution is non- electrolyte ?

THE ANSWER Bec. It doesn't dissociate in water into free ions but spreads among water molecules forming hydrogen bonds



Lesson -2

Classification of solutions according to amount of solute

Classification of Solutions According to Amount of Solute

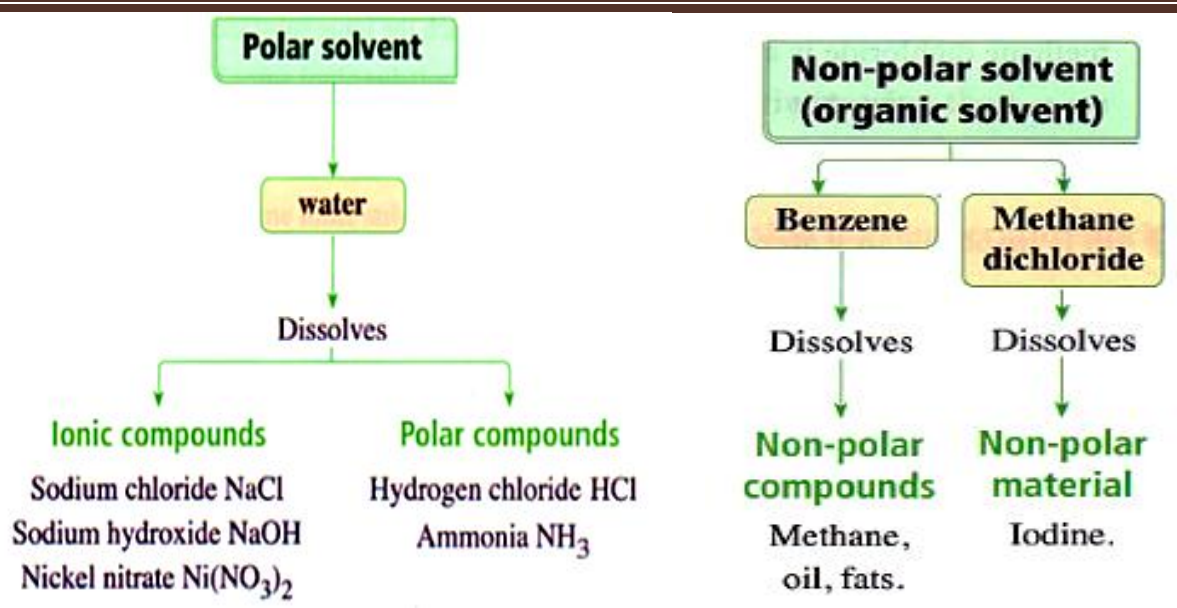
According to the degree of saturation:

1. **Unsaturated solutions:** Solutions that accept dissolving additional amounts of solute at certain temperature
2. **Saturated solutions:** Solutions that don't accept the dissolving of any additional amounts of solute without change in temperature
3. **Super saturated solutions:** Solutions that allow the dissoving of additional amounts of solute with heating.

When super saturated solutions ***cool down***, the ions or molecules of the additional amount of solute gets ***separated*** away from solvent molecules and ***precipitates*** in the form of crystalline solid substance again

Dissolving process:

1. Water molecules are in continuous motion due to their kinetic energy
2. When putting an ionic compound crystal ***(such as NaCl) in water***, water molecules collide with the crystal and attract its ions.
   - ***This process is the dissolution process***



- Polar and ionic compounds dissolve easily in water(polar in polar)
- Non-polar molecules (fats, oil, methane…etc.) don't dissolve in water (but dissolve in benzene).(non-polar in non-polar)
- Dissolution process can be controlled by surface area, stirring and heat

- ***How does oil dissolve in benzene***

- Both of oil and benzene are non-polar compounds whose intermolecular forces are weak, allowing the spreading of oil or benzene molecules

- **Solubility**: the ability of the solute to dissolve in a certain solvent

Degree of solubility is the mass of solute dissolving in **100g** of solvent to form saturated solution at certain temp.

➢ Factors affecting solubility

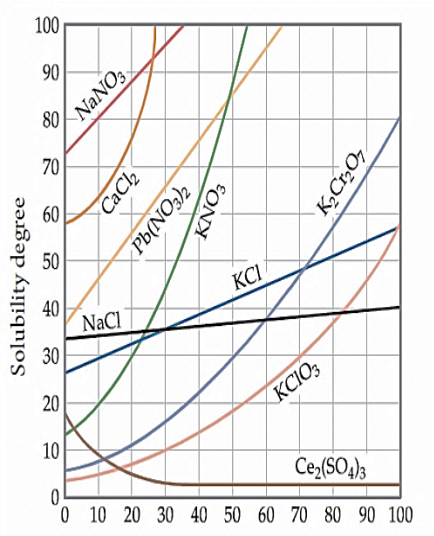
1. The nature of solute and solvent :like in like

The solute dissolves more in similar polarity solvent

- Ammonium nitrate **polar** dissolves in water **polar** larger than in ethanol
- Mercuric chloride **nonpolar** dissolves in ethanol **nonpolar** larger than in water

2. The temperature

- ✓ Solubility of most solids increases by raising the temperature such as: KCl ,KNO_3
- ✓ Solubility of some salts has a weak change increases by raising the temperature such as: ***NaCl***
- ✓ Solubility of few salts decreases by raising the temperature such as : $Ce_2(SO_4)_3$
- ✓ The solubility of NaCl is: 36.8 gm at 25°C and 41 gm at temperature of 80°C



The relationship between solubility and temperatur

- ✓ When dissolving 36.8g of sodium chloride in 100g water at temperature of 25◦C the solvent becomes saturated.

Concentration of solutions

It's said that a solution is "**concentrated**" if the quantity of solute is greater than usual (But not greater that the quantity of solvent)

It's said that a solution is "**diluted**" if the quantity of solute is less than usual.

- ➢ **The concentration of solutions the methods of calculation of**

1- Percent concentration 2- Molarity (M) 3- molality (m)

Colligative (collective) properties of solutions

The properties of solutions differ from the properties of the solvents, forming them after dissolving non-volatiled substances at the same conditions, these new properties are called collective properties.

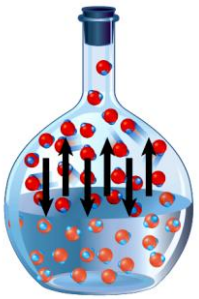
Colligative (collective) properties of solutions : They are properties whose values depend only on the number of solute particles per unit volume of solution and not on the type of solute.

Properties of solutions

Some *of the properties of pure solvent are different from those of solutions. Such as vapour pressure, freezing point and boiling point*

Vapour pressure: The pressure exerted by vapour when it becomes in state of equilibrium with the liquid inside a closed container at constant temperature and pressure

➢ **At state of equilibrium:**



✓ The rate of condensation equals the rate of evaporation inside a closed container.

Vapour pressure: depends on the temperature of the liquid (when the temperature increases, both of the evaporation rate and vapour pressure increase

✓ When vapour pressure becomes equal to the atmospheric pressure, the solution begins boiling "which is known as natural boiling point"

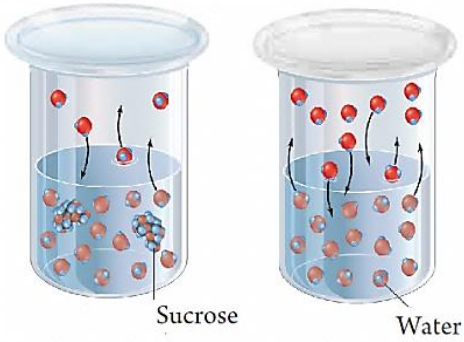
Natural boiling point can prove the purity of liquids

Boiling point: The temperature at which the vapour pressure of solution becomes equal to the atmospheric pressure

- The boiling point of pure water is 100◦C.

G.R: It increases when nonvolatile solute is dissolved in water such as salt?

- Because the solute particles decrease the number of water molecules at the surface ,so the vapour pressure decreases.
  - The salt molecules are attached to water molecules by attraction forces.



Vapor pressure of pure solvent is greater than that of solution contains non volatile solute

- This decreases vapor pressure Therefore; water molecules need more energy which increases the boiling point.
  - The same occurs to any nonvolatile solute in any solvent

G.R. the boiling point of salty water is greater than fresh water

Freezing point:

The temperature at which the substance changes from liquid state into solid state.

- The freezing point of pure water is 0◦C.

G.R:The freezing point decreases when nonvolatile solute is dissolved in water such as salt?

- Because the solute particles are attached to water molecules by attraction forces.
- This keeps the water molecules away from each other and need losing more of energy and this decreases freezing point.
- Salts are put on roads in winter to prevent them from freezing, which decreases the rate of the occurrence of car accidents

Suspensions

- They are non-homogenous mixtures whose solid particles can be precipitated and can be distinguished by naked eyes or microscope
- (the solute and solvent don't mix).
- The diameter of each precipitated solid particle equals 100nm.
- The components of suspensions can be separated using "filter papers".
- The process of separating between the components of suspensions is known as "Filtration process"

Examples: 1-Water and oil 2-Water and chalk 3- Water and sand

G .R :Sugar dissolves in water forming a solution, but milk powder disperses in water?

- Because the size particles of sugar after dissolving is < 1 nm, but that in case of milk powder ranges between 1: 1000 nm.

G .R:Chalk's powder forms a suspension in water?

- Because the size of chalk's particles is > 1000 nm.

- **Colloids:** They are non homogeneous mixtures

- The diameter of particles range between (1:100 n.m.).
- The substance that forms the colloidal particles is called the ***dispersed phase*** the medium in which colloidal particles are dispersed is called **dispersing medium.**
- The colloids can be distinguished from suspensions by using light which is dispersed by colloids as Tindal phenomena

Dispersed phase	Dispersed medium	Examples
Gas	Liquid	Some types of creams – whipped egg.
	Solid	Sweet made of sugar and egg white.
Liquid	Gas	Aerosols – fog.
	Liquid	Milk – mayonnaise – emulsion of oil and vinegar.
	Solid	Hair gel.
Solid	Gas	Dust in air particles.
	Liquid	Pigment – blood – starch in hot water.

- The properties of colloids differ from the real solutions and the suspensions.

1. As many of them, at concentration, take the form of milk or clouds
2. It appears clear at severe dilution
3. The ultrafiltration membrane does not hold its particles
4. If left for a while without shaking, it does not permeate at the bottom of the solution.

➢ Preparation Methods for Colloids

- **Dispersion method:** The substance is crushed into small particles until its size reaches the size of colloid particles, then added to the dispersing medium with stirring as (starch in water).
- **Condensation method:** The small particles are collected together into larger and more appropriate particles by chemical reactions like oxidization, reduction, or hydrolysis.
  - $2H_2S_{(aq)} + SO_{2(g)} \rightarrow 3S_{(colloid)} + 2H_2O_{(l)}$ (oxidation)

Lesson -3
Acids and Bases

Acids and bases play an important role in life.

Acids are used in the industry of fertilizers, explosives, medicines, plastic, car motors, computers...etc.

Bases are used in the industry of soap, detergents, medicines, dyes...etc

The following table shows some chemical and natural products and the acids or bases forming them

Product	The acid
Acidic plants (lemon -oranges - tomatoes)	Citric acid - Ascorbic acid
Dairy products (milk - cheese - yoghurt)	Lactic acid.
Soft drinks	Carbonic acid - Phosphoric acid.

Product	The base
Soap	Sodium hydroxide
Baking soda	Sodium bicarbonate
Washing soda	Hydrated sodium carbonate

Acid: A Compound that has sour taste, turns the color of blue litmus paper into red

1- It reacts with metals forming hydrogen

$$Zn(s) + HCl_{(aq)} \rightarrow ZnCl_{2(aq)} + H_{2(g)}$$

2- It reacts with alkalis (bases) forming salt and water

$$Na_2CO_3(s) + HCl(aq) \rightarrow NaCl\ (aq) + H_2O(l) + CO_2(g)$$

1. *3- It reacts with Carbonate or Bicarbonate salts forming effervescence and* producing carbon dioxide gas CO_2 which makes limewater turbid.

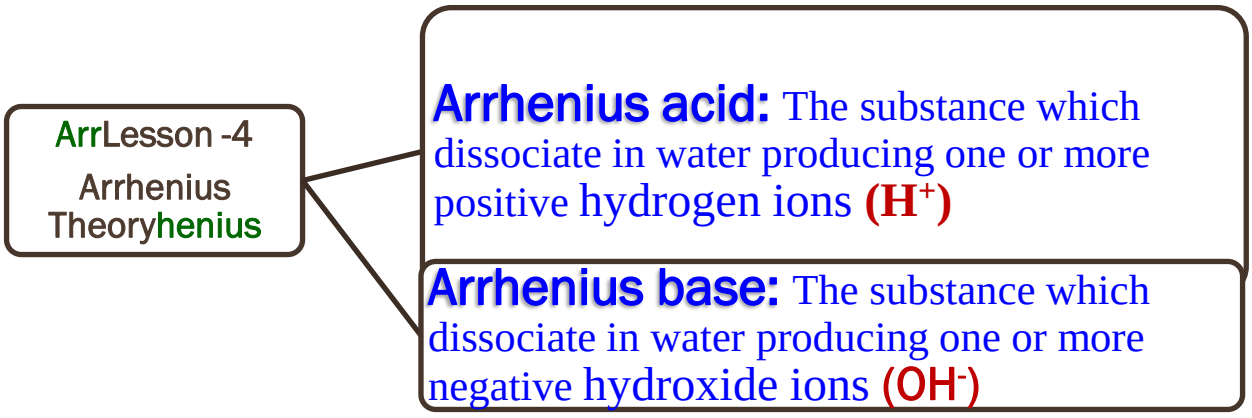
$$Na_2CO_{3(s)} + H_2SO_{4(aq)} \rightarrow Na_2SO_{4(aq)} + H_2O_{(l)} + CO_{2(g)}$$

Sodium carbonate+ Sulphuric acid → Sodium sulphate+ Water+ Carbon dioxide gas

Base (Alkali)

A compound with bitter taste and soap texture which turns color of litmus paper into blue

It reacts with acids forming salt and water



In this theory

- Acids should contain hydrogen atoms and bases should contain hydroxide group

- "Neutralization reactions" the reaction of acids with bases forming water and salts

$$HCl + NaOH \longrightarrow NaCl + H_2O$$

$$H^+_{(aq)} + \cancel{Cl^-_{(aq)}} + \cancel{Na^+_{(aq)}} + OH^-_{(aq)} \longrightarrow \cancel{Na^+_{(aq)}} + \cancel{Cl^-_{(aq)}} + H_2O_{(l)}$$

$$H^+_{(aq)} + OH^-_{(aq)} \longrightarrow H_2O_{(l)}$$

- ✓ Polar water molecule contains oxygen atom carrying (δ-) charge,
- ✓ While hydrogen atoms carry (δ+) charge. So water molecule will be affected by the ions in the aqueous solution one way or another.

Scientists discovered recently that protons (positive hydrogen ions) can't exist freely in aqueous because they bond with water molecules forming solutions (H_3O^+) ions hydronium

- Carbon dioxide forms acidic solutions although they don't contain any hydrogen positive ions in it.
- Some bases **(such as ammonia)** form alkaline solutions although they don't contain any hydroxide ions in them

Lesson -4
Classification of Acids and Bases

Classification of acids

(1)According to strength :

Strong

It is completely ionized in aqueous solutions

HCl - HNO_3 - H_2SO_4

They are good conductors (strong electrolytes)

Weak

It is partially ionized in aqueous solutions

- Acetic acid CH_3COOH
- carbonic acid H_2CO_3

They are bad conductors

✓ There is no relation between the no. of hydrogen atoms and strength of the acid so Nitric acid HNO_3 is stronger than H_3PO_4

According to the nature of their origin:

Organic acids

They are produced from plants or animal origin .

Acetic acid - formic acid - Citric acid - Oxalic acid – lactic acid

Mineral acids:

They contain nonmetal atoms such as Cl, N, S

They haven't organic source

Hydrochloric HCl, Nitric acid HNO_3- Sulphuric acid H_2SO

According to number of basicity

Basicity Number of protons (H^+) which the acid gives in its aqueous solutions

monobasic	Dibasic	tribasic
Produces one proton in aqueous solution	Produces two protons in aqueous solution	Produces three protons in aqueous solution
HCl hydrochloric CH_3COOH acetic	Sulphuric H_2SO_4 Oxalic acid	Phosphoric H_3PO_4 Citric acid $C_6H_8O_7$

Classification of bases :

1. *according to strength :*

1) *strong bases:* it is completely ionized in aqueous solutions

- They are good conductors (strong electrolytes)
- Such as sodium hydroxide (NaOH) - potassium hydroxide (KOH)

2) Weak bases : partially (incompletely)ionized in aqueous solutions

- ✓ They are bad conductors (weak electrolytes)
- ✓ Such as ammonium hydroxide NH_4OH

2- according to its molecular composition :

Metal oxides:-	Metal hydroxides:-
1- Iron oxide (FeO)	1- Potassium Hydroxide (KOH)
2- lead oxide (PbO)	2- Sodium hydroxide (NaOH)
6- Potassium oxide (K_2O)	3- 4- Magnesium hydroxide $Mg(OH)_2$

Metal carbonates (or bicarbonates):-

1- Sodium bicarbonate ($NaHCO_3$)

4- Potassium carbonate (K_2CO_3)

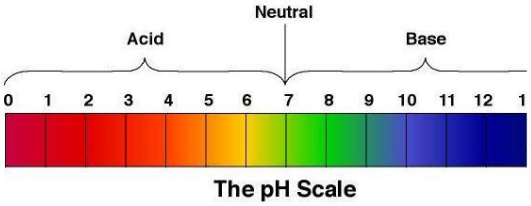
- ➢ Bases which dissolve in water producing negative hydroxide ions are called "Alkalis"
- ➢ We can say that all alkalis are bases but all bases are not alkalis

Detection of acids and bases

Indicator: *weak acids or bases whose colors change by the change of the type of solution*

The indicator	In acids	In bases	neutral solutions
Methyl orange	Red	Yellow	Orange

Bromothymolblue	Yellow	Blue	Green
Phenolphthalein	Colorless	Fuchsia	Colorless
Litmus paper	Red	Blue	Purple



- **Hydrogen power P^H:**
- There are methods for determining the type of solution in numerical values ranging from 0 to 14.
  - Papers and digital devices are used to measure the pH of solutions

The reaction of metals with diluted acids:

- Metals which come before hydrogen in chemical activity series replace it in its diluted acids forming salt and hydrogen gas
  - Active metal + diluted acid ⟶ Salt of acid + Hydrogen

$$Zn_{(s)} + H_2SO_4 \longrightarrow ZnSO_{4(aq)} + H_{2\,(g)}$$

- *The reaction of metal oxides with acids:*

Metal oxide + Acid ⟶ water + salt of acid

$$CuO_{(s)} + H_2SO_4 \longrightarrow CuSO_{4\,(aq)} + H_2O_{(l)}$$

- *The reaction of metal hydroxide with acids:*

Acid + Alkali ⟶ Salt + Water

$$HCl_{\,(aq)} + NaOH_{\,(aq)} \longrightarrow NaCl_{(aq)} + H_2O_{\,(l)}$$

Neutralization: reaction of acid with alkali forming salt and water

They are used in chemical analysis process in order to calculate the concentration of an acid or alkali using in the presence of suitable indicators.

The reaction of metal carbonates (or bicarbonates) with acids

- ✓ This reaction is used to test acidity
- ✓ $Na_2CO_{3(s)} + 2HCl_{(aq)} \longrightarrow 2NaCl_{(aq)} + H_2O_{(l)} + CO_{2(g)}$

Nomenclature of salts: [Anion: Negative ion - Cation: Positive ion]

- ✓ Some acids have two kinds of salts because their molecules have more than one hydrogen atom as dibasic acids (such as sulphuric and carbonic acids) OR Tribasic acids (such as phosphoric acid) have three kinds of salts because their molecules contain 3 hydrogen atoms
- ✓ The names of salts which contain hydrogen in their anions are called "bi_" "Hydrogen "such as HSO_4 is called "Bisulphate" or "Hydrogensulphate"
- ✓ In the case of organic acids (such as acetic acid) the anion of acid is written on left CH_3COOK (such as Potassium acetate)

Solubility of salts

Salt	Acid	Base	Type of salt	PH
(NH_4Cl)	Strong	Weak	acidic	PH < 7

K_2CO_3 Or Na_2CO_3	Weak	Strong	basic	PH > 7
CH_3COONH_4	Weak	Weak	neutral	PH = 7
NaCl	Strong	Strong	neutral	PH = 7

Lesson-5 Evluation

First : Choose the correct answer :

1. Air represents a gaseous solution of the ..... type.

A. Gas in gas B. Gas in liquid

C. Liquid in gas D. Solid in gas

2. Water is polar solvent due to the difference in electronegativity between oxygen and hydrogen and the angle between the bonds whose value is about .....

A. 104.5° B. 105.4°

C. 90° D. 140.5°

3. ..... is an example of strong electrolytes.

A. $H_2O_{(\ell)}$ B. Benzene

C. $HCl_{(g)}$ D. $HCl_{(aq)}$

4. The unit used for expressing the molar concentration of a solution is .....

A. mol/L B. g/eq.L

C. g/L D. mol/kg

First : Choose the correct answer :

1. The phosphoric acid H_3PO_4 is one of the ..... acids

   A. Mono-proton B. Di-proton

   C. Tri-proton D. Poly-proton

2. The pH value of an acidic solution is .....

   A. 7 B. 5

   C. 9 D. 14

3. In the reaction of ammonia with the hydrochloric acid, the ammonium ion $(NH_4)^+$ is cons .....

   A. Conjugated acid B. Base

   C. Conjugated base D. Acid

4. One of the following acids is considered a strong acid .....

   A. Acetic acid B. Carbonic acid

   C. Nitric acid D. Citric acid

5. The pH value in which the color of phenolphthalein is red is .....

   A. 2 B. 4

   C. 6 D. 9

6. The conjugated acid HSO_4^- is .....

   A. HSO_4^+ B. SO_4^{-2}

   C. H_2SO_4 D. H^+

1. In the neutral median , the indicator that has a purple color is .....

A. Litmus
B. Phenolphthalein
C. Orange methyl
D. Blue bromothynol

2. The pH number for a base solution .....

A. 7
B. 5
C. 2
D. 8

3. Acids react with carbonate and bicarbonate salts and ..... gas rises

A. Hydrogen
B. Oxygen
C. Carbon dioxide
D. Di-sulfur oxide

4. Dissolve 20 g of sodium hydroxide in an amount of water and complete the solution until 250 ml, the concentration is ..... [Na = 23 , O = 16 , H = 1]

A. 1 M
B. 0.5 M
C. 2 M
D. 0.25 M

5. All of the following are mineral acids except .....

A. Sulfuric acid
B. Citric acid
C. Phosphoric acid
D. Hydrochloric acid

6. Which of the following salts form a solution of an alkaline affect on the litmus? .....

A. NH_4Cl
B. K_2CO_3
C. $NaNO_3$
D. KCl

7. If one mole is dissolved from each of the following substances in one liter of water, which one of them has the largest effect in the vapor pressure of its solution? .....

A. KBr
B. $C_6H_{12}O_6$
C. $MgCl_2$
D. $CaSO_4$

Unit Three: Types of Bonds

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Unit Three: Types of Chemical Bonds

Dear student by the end of the following unit you should have the following abilities and knowlages:

The abilities: To be able to:

1-Distinguish between different types of bonds,
2- Distinguish between properties of covalent and ionic compounds.
3- Predict the different types of bonds between elements,
4- Differentiate between hydrogen and covalent compounds.
4-Explain the different between metals in their electrical conductivity.

Knowledge:To learn about :

1-Theories that explain the covalent bonds,

2-Properties of both covalent and ionic compounds ,

3-Types of chemical and physical bonds.

Lesson -1
Ionic and covalent bonds

Types of Bonds

Chemical combination

The chemical reactions depend only on the electrons of outermost energy level.

- *Write electronic configuration of the following elements:*
- $_{11}Na$ .....
- $_{10}Ne$.....
- The elements are two types active and inactive "inert".
- **(Inert gases are chemically inactive).**

العنصر	التوزيع الالكتروني
$_{2}He$	2
$_{10}Ne$	2 , 8
$_{18}Ar$	2 ,8 ,8
$_{36}Kr$	2 , 8 , 18 ,8
$_{54}Xe$	2 , 8 , 18 ,18,8
$_{86}Rn$	2 , 8 , 18 ,18,32 ,8

This is due to their outer energy levels are completely filled with electrons (stable elements).

(The other elements are chemically active).

- Bec.Their outer energy levels aren't completely filled with electrons (unstable elements) so they try to complete them in the chemical reactions to be as stable as inert gases.

G.R.: helium doesn't form compounds while hydrogen does

.....

.....

G . R :On mixing iron fillings with sulphur the result will be a mixture not a compound.

- Bec. There is no chemical bond formed between iron and sulphur and each substance has its own properties.
- ***But if this mixture is heated enough to form new bond the result will be a compound called iron sulphide FeS.***

Fe + S → FeS

*There is a bond is formed in**FeS**so this is chemical reaction*

Chemical reaction:*Is the process of breaking bonds in reactants are forming new bonds in products*

The chemical reactivity of the atom depends on outer level

The electrons of outer level "valence electron" act in forming bonds and its nature.

There are two types of bonds:

1- Chemical *2- physical*

Chemical bonds

1. **Ionic bond:**

The elements are :

- **Metals** contains 1,2 or 3 electrons (less than its half capacity)in outer most energy level such as sodium $_{11}$Na "2,8,1".
- **Nonmetals** contain greater than its half capacity such as chlorine $_{17}$Cl "2,8,7".

The metal atom loses electrons and form cations (+ve ion

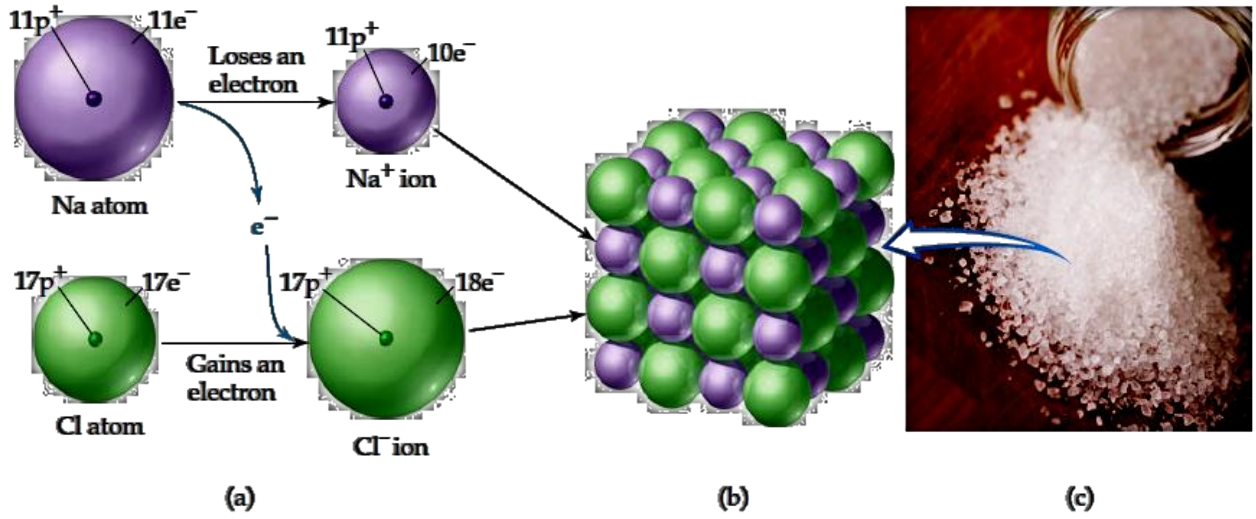
This is due to:

1- Large volume of metals' atoms (radius) and Small ionization potential

The nonmetal atom gains these electrons and form anions (-ve ion)

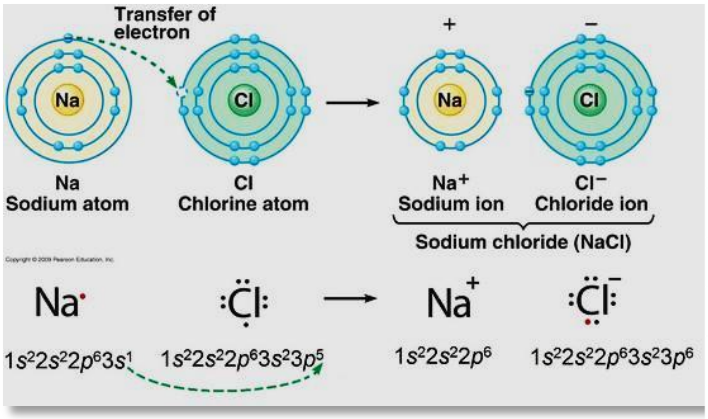
This is due to:

1- small volume of metals' atoms (radius) and great electron affinity , both atoms after this have the electron configuration of the nearest inert gas and become stable



Formation of an ionic compound.

(a)The transfer of an electron from Na atom to Cl atom leads to the formation of Na^{+} ion and C^{l-} ion.



(b)Arrangement of these ions in solid crystalline form sodium chloride, NaCl.

↳ Electrostatic attraction occurs between (+ve) cations & (-ve) anions. It is called the **ionic bond.** As shown:

- This means that ionic bond has **no** materialistic existence.

The ionic bond is formed between **metal** and **non-metal** atoms

↳ The difference in electronegativity between the two atoms must at least =1.7

↳ The **strength** of ionic bond is directly proportional to the **difference** in electronegativity between the two ions' atoms.

	Na	***Mg***	***Al***	***Cl***
Electro negativity	0.9	1.2	1.5	3
Different in electronegativity	NaCl 3-0.9=2.1	$MgCl_2$ 3-1.2=1.8	$AlCl_3$ 3-1.5=1.5	
Melting point (C)	810	714	190	
Boling point(C)	1465	1412	(sublimes)	
Conductivity of its molten	Very good	Good	Does not	
Type of bond	Ionic	Ionic	covalent	

✓NaCl and $MgCl_2$are ionic while $AlCl_3$ is covalent

General properties of ionic compounds:

1. Their solutions or molten form are Good conductor of electricity
   - The conductivity depends on the degree of ionization and the difference in electronegativity
   - Molten NaCl is better conducting than molten $MgCl_2$
2. Ionic compounds generally have high melting and boiling points

Because a great amount of energy is needed to break down the crystal lattice and overcome the strong electrostatic attraction force between cations and anions.

II- covalent Bond:

✓ It is formed between two atoms of non-metal and nonmetal

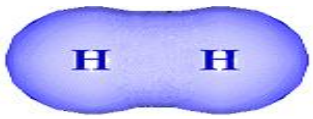
✓ the same element (have the same electro- negativity)

Or between atoms of different elements have closed electronegativity Or the difference in E.N. less than 1.7

✓ It occurs due to sharing of valence electrons

✓ There are three types of covalent bond:

1- Pure covalent:

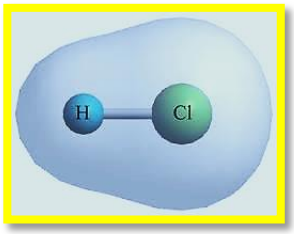


Formed between two similar atoms "has the same E.N" In this case, the two atoms have the same E.N.

Thus, the electron pair spends the **same time** in the vicinity of each atom and the net charge on each atom is zero.

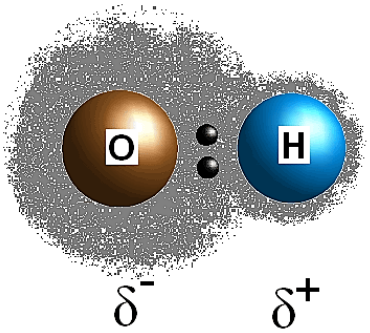
Such as in H-H , O=O , Cl-Cl .

2- Polar covalent:



Formed between two atoms have difference in E.N less than 1.7

Ex: HCl molecule



In this case, because chlorine atom has more E.N., so it has greater ability to attract the pair of electrons of the covalent bond (i.e. the electrons spend more time at a chlorine atom). As a result, chlorine atom acquires a partial negative charge (-δ) and not complete one (as in the case of chloride ion Cl^-), while hydrogen atom acquires a partial positive charge (+δ).

Polar molecules: H_2O , HF and NH_3

3- Non-Polar covalent:

Formed between two atoms have difference in E.N less than 0.4

Ex: H_2S - CH_4 molecules

Electronegativity and Bond Polarity

- We can use the difference in electronegativity between two atoms to gauge the type the formed bond .

	F_2	HF	LiF
Electronegativity difference	4.0 − 4.0 = 0	4.0 − 2.1 = 1.9	4.0 − 1.0 = 3.0
Type of bond	pure covalent	Polar covalent	Ionic

1. In F_2 the electrons are shared equally between the fluorine atoms and, thus, the covalent bond is ***pure***.

2. In HF the fluorine atom has a greater electronegativity than the hydrogen atom,with the result that the electrons are shared unequally—the bond is ***polar***.

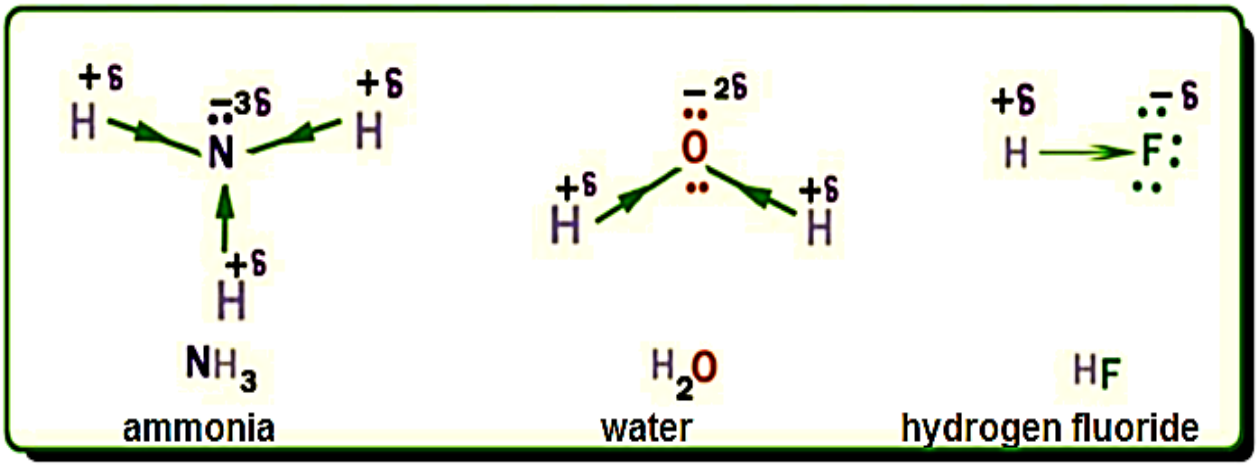
3.In LiF the electronegativity difference is very large, meaning that the electron densityis shifted far toward F. The resultant bond is ***ionic***.

Lesson -2
Theories that Explain the Covalent Bonds.

(1) Electronic theory of valence: (Octet rule Theory)

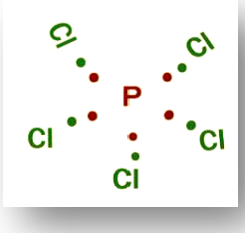
Lewis and Kossel scientist supposed that:

"All atoms of elements have tendency to reach the octet structure of the outer energy level to be as the nearest inert gas expect H, Li and Be

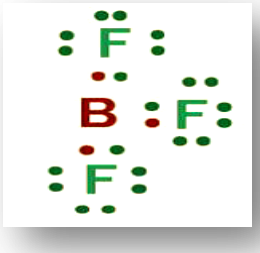


Defects of electronic theory of valence:

1- It failed to explain the binding in many molecules ,Which No. of electrons is around central atom is more or less in which than 8



As In PCl_5:Phosphorus is surrounded by 10 electrons , and BF_3, Boron is surrounded by 6 electrons.



2- It couldn't explain some properties of molecules as stereo structure and angles between bonds.

2-The valence shell electron pairs repulsion theory: VSEPR theory

In chemical compounds there are two types of electron pairs:

1) Bond pair : which forming the bonds
2) Electron pair : which exist on atoms without forming bonds

- The VSEPR model is based on the idea that electron pairs are negatively charged and therefore repel one another.
- The best arrangement of a given number of electron is that minimizes the repulsions among them.

1. Two electron pairs orient linearly,
2. Three orient in a trigonal-planar
3. Four orient tetrahedral shape.

example	Molecular geometry	Arrange of electron pairs	Free pairs	Banded pairs	Total pairs
F-Be-F	AX_2 Linear	linear	0	2	2
BF_3	AX_3 Trigonal planar	Plaaner triangle	0	3	3
$[NO_2]^-$	AX_2E angular Bent		1	2	
CH_4	AX_4 tetrahedron	Tetrahedral	0	4	4
NH_3	AX_3E three-base pyramid		1	3	
H_2O	AX_2E_2 angular Bent		2	2	

Effect of Nonbonding Electrons and Multiple Bonds on Bond Angles

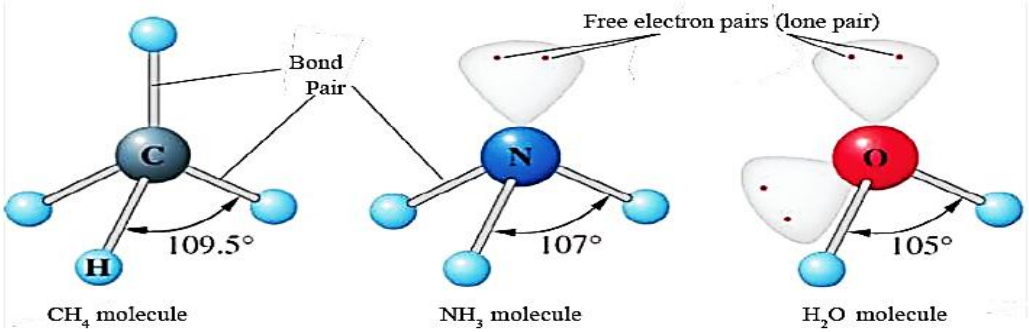
- The lone pair of electrons controls the value of bond angles in the molecule ?
  - Because the lone pair of electrons are bind to nucleus of central atom from one side and spread to the space from other side.

- While the bond pair binds from both sides with the nuclei of the two bonded atoms.
- Increasing the number of lone pair of electrons in central atom of the molecule leads to increasing the repulsive force between them which cause decrease value of angles between covalent bond of molecule and in general, the repulsion is between:

(Lone pair , lone pair) > (Lone pair, bond pair) > (bond pair, bond pair)

How can you explain decreasing the values of covalent bond angles in water than ammonia than methane?

The lone pair of electrons controls the values of bond angles in covalent molecules



Free electron pairs control in determining the values of angles among the bonds in covalent molecules.

In The Ammonia molecule contains 3 bond pairs

The molecule contains one lone pair and involves three covalent bonds.

The stereo structure of the molecule is three-base pyramid and its abbreviation is AX_3E

Q : Conclude the number of bond pairs and lone pairs and then arrangement of the electrons pairs for the molecules which have the following abbreviations :(a)AX_4 b)AX_2E



(a)X = No. of bond pairs= 4 E= No. of lone pairs= 0

- The total no. of the electron pairs = 4 + 0 = 4
- The arrangement of the electron pairs is tetrahedron.

(b)X= No.of bond pairs=2 E =No.of lone pairs =1

- The total no. of the electron pairs = 2 +1 =3.
- The arrangement of the electron pairs is planar triangle.



SO_2 molecule is abbreviated as AX_2E , while H_2O molecule is abbreviated as AX_2E_2 ,although each of them consists of three atoms.

Because the central atom (A) in SO_2 molecule is bonded with The oxygen atoms (O_2) and carries one lone pair of electrons (E), while the central atom (A) in H_20 molecule is bonded with two hydrogen atoms (H_2) and carries two lone pair of electrons (E_2).



G.R. BeF_2 molecule is linear, while SO_2 molecule is angular , although the central atom in each of them is bonded to two atoms.

Because the central atom in BeF_2 molecule doesn't carry any lone pair of electrons, so the repulsion force between the two bond pairs is as large as possible which makes the linear stereo structure of the molecule.

While the central atom in SO_2 molecule carries lone pair of electrons which repulses strongly with the bond pair and this leads to reduce the angle value between them and makes the angular stereo structure of the molecule.

The stereostructure of ammonia molecule is three-base pyramid, while the arrangement of electron pairs in the same molecule takes tetrahedron shape.

Because of bonding the central atom with three bond pairs, while the arrangement of the electron pairs in tetrahedron shape because the total number of electron pairs (bond and lone) = 3 + 1 = 4

Therelation between number of lone pairs of electrons and angles value between the covalent bonds in the molecule.

When the number of lone pairs of electrons in the central atom increases, the repulsion force increases, causing a decrease in the angles value between the covalent bonds in the molecule.

Give reason:

The angle value between the covalent bonds in ammonia molecule is less than that in methane molecule.

Because the central atom in ammonia molecule carries one lone pair of electrons which repulses with the bond pairs and that leads to decreasing the angle value between them.

While the central atom in methane molecule doesn't carry a lone pair of electrons, so the angle value between the bond pairs is larger than that in ammonia molecule.

Melting and boiling points:

Relatively low melting & boiling points due to the very weak attraction force between their molecules.

Give reason: Solution of HCl in benzene does not conduct electricity, but its solution in H_2O is a good conductor of electricity.

Because HCl is a polar covalent compound which can diffuse in benzene (non-polar solvent) but can't be ionized into (+ve) ion (H^+) & (-ve) ion (Cl^-), so it can't conduct electricity.

✓ While HCl in H_2O (polar solvent) can be ionized into ions which can migrate to the opposite electrodes and so conducts electricity.

Lesson -3
The co-ordinate bond

III – Coordination bond

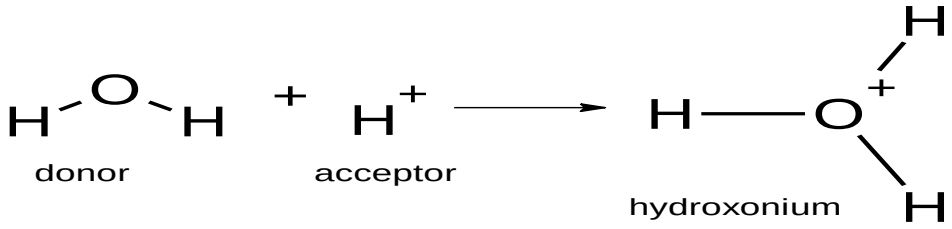
"It is a type of covalent bond formed between 2 atoms on of theme has one Orbital containing **alone pair** of electrons which is called **donor** atom, while the other atom has a **vacant** orbital called **acceptor** atom

- The pair of electrons forming the bond are from one atom.

Example:

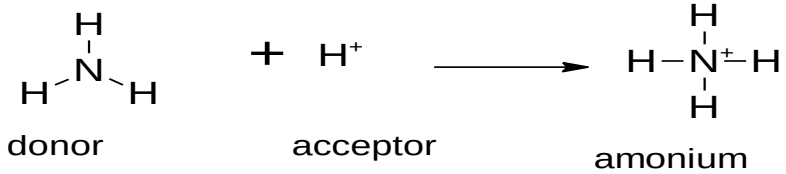
Hydronium ion (hydroxonium) H_3O^+

Is formed when a strong acid dissolved in water:



Give reason: proton of strong acid does not exist freely in water

Ammonium ion $(NH_4)^+$



✓The proton is acceptor while central atom is donor like oxygen in $(H_3O)^+$, phosphorous in $(PH_4)^+$ & nitrogen in $(NH_4)^+$.

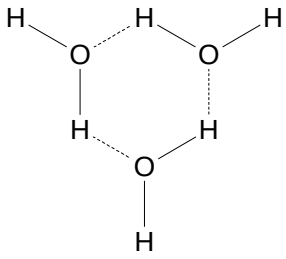
✓the bonds in the last examples are polar covalent and co – ordinate bonds

IV : Hydrogen bond :

- ✓It is a bond formed between polar molecules in which hydrogen atoms lies between to atoms of high electronegativity as (oxygen) or (fluorine)
- ✓The hydrogen atom binds with one atom by polar covalent bond and binds with the second atom by hydrogen bond.
- ✓So hydrogen atom acts as a bridge to bind molecules together.

Explanation of hydrogen bond in water :

(1) oxygen atoms has small volume ,so it has high electronegativity (3.5) , while electronegativity of hydrogen is 2.1



So oxygen atom will carry -δ charge, while hydrogen atom carry +δ charge.

(2) Hydrogen bond is formed due to the **attraction** force between one hydrogen atom of one molecule and one molecule and one oxygen atom of another molecule , so molecule of water are collected by hydrogen bonds , so water exists in a liquid state and has relative high boiling point



Although molecular weight of water (H_2O) is very small (18) but it exists as liquid and boils at 100°C, while molecular weight of (H_2S) is (34°C) but it exists in a gaseous state and boils at (-61C°).

Bec. The difference in electronegativity between **O** and **H** is greater than that between **S** and **H** so the hydrogen bond in water is stronger than that in H_2S

Although sugar is covalent compound but it dissolves in water.

Due to formation of Hydrogen bond between hydroxyl group of sugar & oxygen of H_2O, but its solution is a bad conductor of electricity because it can't be ionized.

1- Properties of hydrogen bond:

(1) Strength of H-Bond depends on the difference in electronegativity as it increases, the strength and the boiling point will increases . As in water.

(2) It's longer than covalent bond.

(3) It's is much weaker than covalent bond.

(4) It's has several forms:

1.Straight line. B- Closed ring. C- Open net.

V- Metallic Bond (between atoms of metal in the metallic structure):

"Is formed from electron cloud of the free valence electrons around (+ve) metal ions."

The free valence electrons of the outer shell are associated together forming an electron cloud which decreases the repulsion force between (+ve) ions in the metallic structure. The strength of the

metallic bond depends on no of free valance electrons. As the No. of free valence electrons increases, the atoms of metal will be strongly bonded, so the metal will be harder, of higher melting & boiling points and higher thermal and electrical conductivity.

G.R :The elements of group IA as Na are soft and have low melting point while elements of group IIIA as AI are hard and have high melting point?

Answer: In case of Na weak metallic bond one electron while in case of Al: due to strong metallic bond three valence electrons

The metallic bond strength

The number of valence electrons in the metal atom plays an important role in the strength of the

- metallic bond. As the number of valence electrons increases in the metal atoms, the atoms become more strongly bound and accordingly the metal becomes harder and has a higher melting point.

- This is an evidence when comparing the properties of sodium, magnesium and aluminum (the metal of the third period), as the following table :

	Sodium $_{11}$Na	**Magnesium $_{12}$Mg**	**Aluminum $_{13}$Al**
The electron configuration	$[_{10}Ne], 3s^1$	$[_{10}Ne], 3s^2$	$[_{10}Ne] , 3s^2 , 3p^1$
Number of the outermost electrons	1	2	3
Hardness (Moh's scale)	 Soft enough to be cut by a knife (0.5)	 Mild enough to be bend (2.5)	 Hard easily moulding (2.75)
Melting point °C	98°C	650°C	660°C

Lesson 4
Evaluation

QUESTION 1:-

What is meant by:

1- Chemical reaction

2- Ionic bond

3- Covalent bond

4- Coordinate bond

5- hydrogen bond

QUESTION 2:-

Choose the correct answer:

1- Three elements of atomic numbers C_{11}, B_{10}, A_9: the possible combination takes place between the atoms:

a- B with C b- A with B

c- b with it self d- C with A

2- When two atoms of an element of atomic no 9 combine together to form a molecule, the formed bond is:

a- metallic b- coordinate c- ionic d- covalent

3- The bond in hydrogen fluoride molecule is polar covalent, because the atoms are different in :

a- location in periodic table

b- Electron affinity

c- Electronegativity

d- Ionization potential

When two oxygen atoms combine together to form a molecule , this process takes place by :

a- each atom shares by one electron to form one covalent bond.

b- one atom donates a lone pair of electrons to the other one .

c- each atom shares by two electrons.

d- a polar covalent bond is formed between the two atoms

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Unit Four: Electrochemistry

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Dear student by the end of the following unit you should have the following abilities and knowlages:

The abilities: To be able to:

1-Distinguish between galvanic and electrolytic cells.

2- Distinguish between first and second galvanic cells.

3-Protect the elements against corrosion.

4- Form a simple electrolytic cell for plating metals

Knowledge:To learn about :

1- Getting electric current from chemical reactions.

2- Electrode potential for different elements

3- Extraction , purification of different elements

Lesson one: The galvanic cell

Electrochemistry

Introduction:

Electric energy is considered as the most important and safe form of energy for environment. Electrochemistry is interested to study the exchange conversion of chemical energy and electrical energy through oxidation and reduction reactions. In these reactions the electrons are transferred from one of the reacting substances to the other during the chemical reaction. The following experiment represents one of oxidation – reduction reaction.

Experiment:-

Dip a Zn-sheet in blue copper sulphate solution, Fig .(4 -1a).

You will notice red copper metal precipitated on the surface of zinc sheet, while Zn-metal dissolves in the solution, fig (4-1b).

If this is continued for long period, the blue colour of copper sulphate solution decreases and becomes colourless and the dissolution of (Zn) increases.

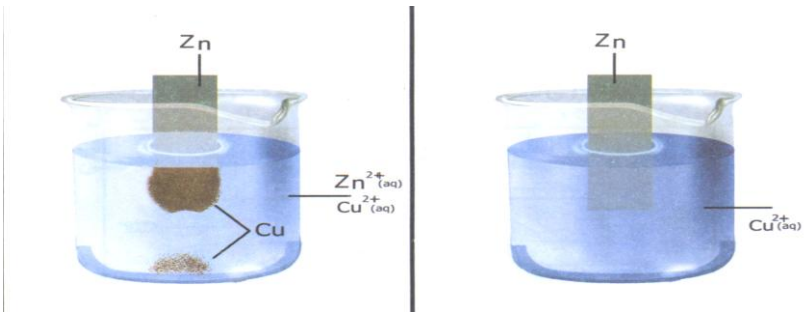
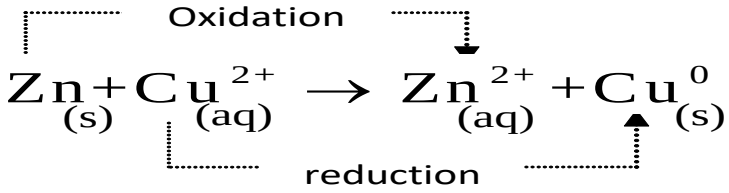


Fig :4-1b

Fig :4-1a

This reaction is spontaneous redox– reaction and can be expressed by the following equation:



We observe that this reaction consists of two half reactions:

Oxidation reaction : $Zn^{0}_{(s)} \longrightarrow Zn^{2+}_{(aq)} + 2e^-$

In this reaction Zinc– atom (Zn^0) losses two electrons and converts to zinc ion (Zn^{2+}) dissolves and defused in the solution.

Reduction reaction : $Cu^{2+}_{(aq)} + 2e^- \rightarrow Cu^{0}_{(s)} \downarrow$

In this reaction copper ion (Cu^{2+}) in the electrolytic solution accepts two electrons (come from the Zn – half cell) and converts to copper metal (Cu^0) deposited on the surface of the Zn- sheet.

Scientists had succeeded in arranging a system known by galvanic cells consists of two half-cell their contents are found in isolated containers connected together by a salt bridge, the electrons are allowed to pass through an electric wire between the half cells to obtain electric current produced from the spontaneous oxidation – reduction reaction which happens in the galvanic cell.

There is another kind of electric cells in which the electrical energy from an external source (dry cell) used to form oxidation – reduction reactions (nonspontaneous) .this kind of cells is known as electrolytic cells or electrolysis cell.

1- Galvanic cells

This is the kind of electric cells from which we can obtain electric current as a result of spontaneous oxidation – reduction reaction like the Daniell cell , shown in Fig .(4-2)

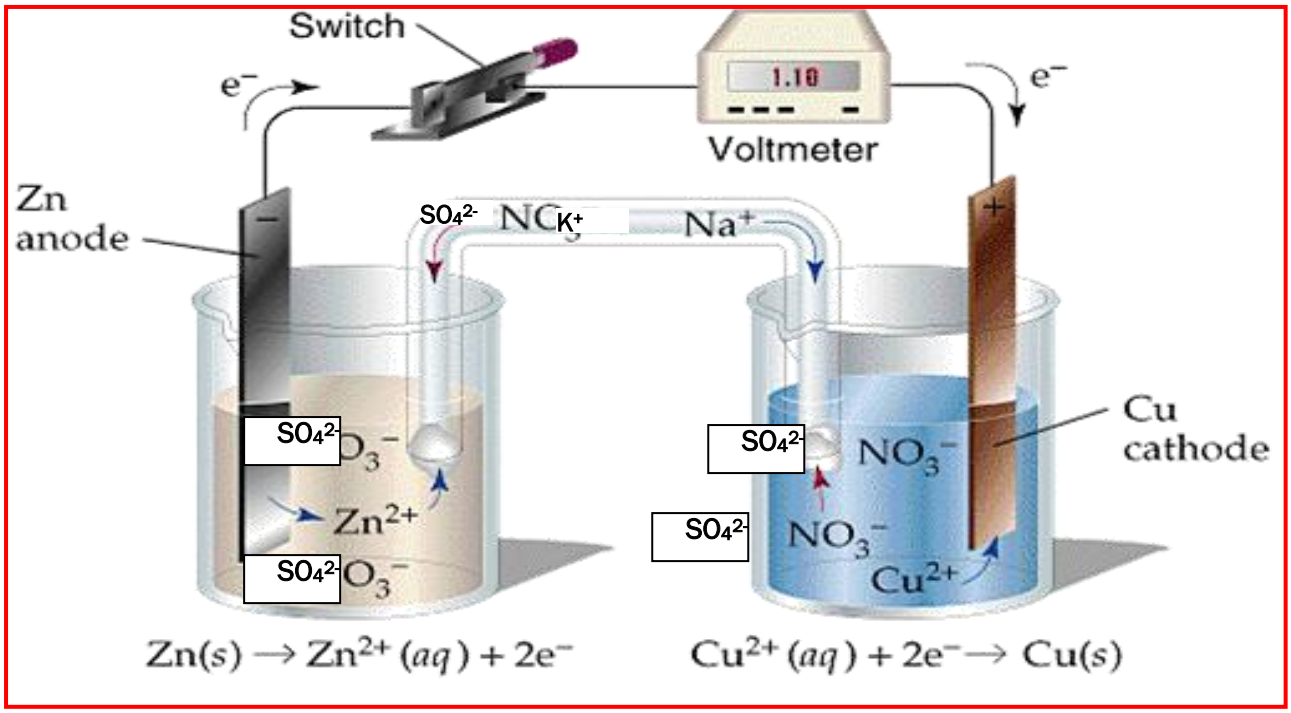


Fig. (4-2)

This cell consists of copper plate (cu-electrode) and zinc plate (zinc electrode) each of them is dipped in one its salt solutions in two separate containers and they are connected together by a salt bridge. Copper plate is the cathode and it is the positive pole of the cell, while the zinc plate is the anode and it is the negative pole of the cell – the solution in each half cell is known as the electroyte.

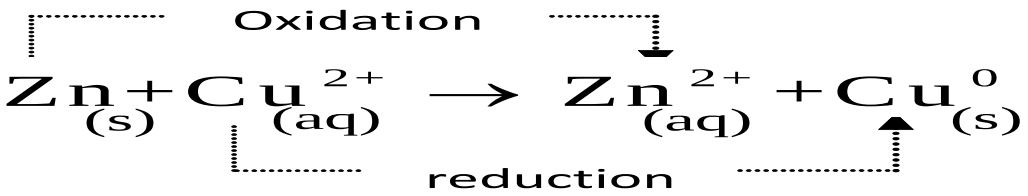
When the two electrodes of the cell are connected by a metal wire the electric current will pass.

The oxidation – reduction reaction in this cell can be explained as follows:

A- At anode: $Zn^{o}_{(s)} \rightarrow Zn^{2+}_{(aq)} + 2e^{-}$ oxidation

B- At cathod $Cu^{2+}_{(aq)} + 2e^{-} \rightarrow Cu^{o}_{(s)} \downarrow$ reduction

Total cell reaction occurring in cell Fig (4-2) is the sum of the two half- cell reactions.



The flow of current is stopped between the two half cells when all of zinc metal dissolves in the zinc half-cell or all the copper ions disappear because they are precipitated as a copper atoms (Cu^{o}) in the copper half cell .

You should notice that oxidation process occurs always at the anode while the reduction process occurs at the cathode in all electric cells.

*What is a salt bridge? What is its role in galvanic cells?

The salt bridge in galvanic cells like (Daniel cell) is a glass U- shaped tube filled with strong electrolytic solution like (Na_2SO_4) its ions do not react with the ions present in the two half cells as well as with the electrode materials of the galvanic cell.

The salt bridge connects between the solutions of the two half cells indirectly. Also it neutralize the excess of both the positive and negative ions which forms in the solutions of the two half cells as a result of the oxidation and reduction reactions in zinc and copper half cells respectively. The absence of the salt bridge in the galvanic cell lead to stopping the oxidation and reduction reaction and

consequently stopping the flow of electric current in the external wire connecting between the two half cells .

Measurement of Electrode Potentials

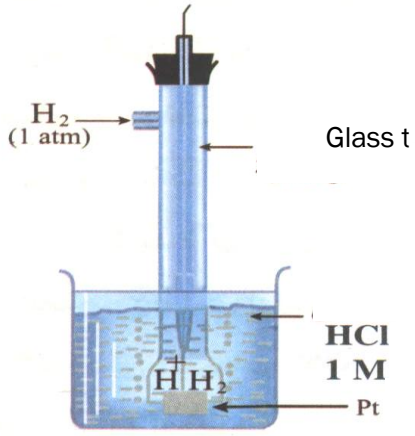


Fig :4-3

There is not certain and direct way for measuring the absolute electric potential difference between metal electrode and solution of ions in galvanic cells while the difference between the two electrode potentials of the galvanic cell can be easily measured. This is by forming of galvanic cell from two electrodes, one of them is the electrode that required to measure its potential and the other is standard electrode which has a constant known potential. Then measure the electromotive force of the cell (cell potential) is measured and from it the unknown electrode potential can be calculated.

Scientists had agreed to use the standard hydrogen electrode as a standard electrode; the potentials of other electrodes can be measured. by knowing its potential equals **zero** volt,

The standard hydrogen electrode as shown in fig (4-3) consists of a platinum sheet ($1cm^2$) covered by a layer of black spongy platinum over which a flow of hydrogen gas at constant pressure of one atmosphere passes and dipped in a molar concentration (1M) of any strong acid solution .

The hydrogen electrode at such conditions is known as the standard hydrogen electrode (SHE) its potential = zero volt.

The potential of this electrode changes by changing the hydrogen ion concentration in the solution or by changing the partial pressure of the hydrogen gas or both.

The standard hydrogen half-cell is represented by a diagram

$Pt - H_2$ (1 atm.) / $2H^+$ (1mol/L)

The ELECTROMOTIVE SERIES

Scientists had measured the standard electrode potentials (E^0) for all metals and nonmetals half cells relative to the standard hydrogen electrode.

If these standard potentials are descending arranged with respect to the –ve reduction potentials and ascending relative to the positive reduction potentials, in which the more negative values are in the upper of the series and the more positive values are lower, we obtain what is known the electromotive series of elements. Which declares the following facts?

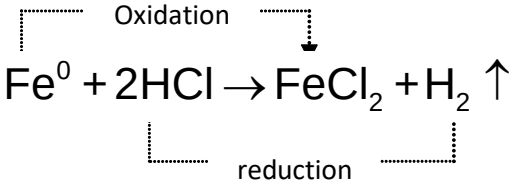
A- The elements which have the more negative values of reduction potentials and found at the top of this series are strong reducing agents and oxidized easily when they react with ions of any element in a lower position in the electromotive series.

B- The elements which have the more positive reduction potential which lie down in the series are strong oxidizing agents (when the metals are in ionic forms and the non-metals are in their elemental state), like fluorine molecules (F_2) . This means that it has more ability to accept electrons when it reacts with any element lies above it in the electromotive series.

C- The elements lie above in the series replace the elements which lie down in its salt solution and as the distance increase between two elements in series , the ability of the pre element (of most negative potential, or lower positive) to expel the following element (of lower negative potential or most positive) from its compounds .

D- All elements lie above hydrogen in the electromotive series replace hydrogen ions in acidic solutions (i. e hydrogen gas evolves).

Example : reaction of iron metal with hydrochloric acid



As the negative potential value increase, the ability to substitute hydrogen ions increases, while the positive potential elements which lie below hydrogen in the electromotive series do not replace hydrogen ions in its solutions.

It is interesting to refer her that the standard half cell potential for any element takes negative or positive sign. If the reaction in zinc half cell as example is a reduction process, the potential here is known as the standard reduction potential (E^0)

$Zn^{2+} + 2e^- \rightarrow Zn^0$ (reduction process)

And the **reduction potential** (E^o) of zinc half cell is pre estimated relative to the standard hydrogen potential as is found to equal – 0.76 volt , but the **oxidation potential** to this element has opposite sign since it represents here oxidation process .

$$Zn^o \longrightarrow Zn^{2+} + 2e^- \qquad E^o = +0.76\ v$$

Lesson two: The electrode potential series

Although it is also possible to arrange the electromotive series of elements according to the standard oxidation potentials in descending order relative to the positive potentials and ascending relative to the negative potentials. as shown in table in page (82)

The ELECTROMOTIVE SERIES of Elements

Half cell Half reaction	Standard oxidation potential (volt)	Standard reduction potential (volt)
$Li \rightleftharpoons Li^+ + e$	+3.045	- 3.045
$K \rightleftharpoons K^+ + e^-$	+2.924	-2.924
$Na \rightleftharpoons Na^+ + e^-$	+2.711	-2. 71 1
$Mg \rightleftharpoons Mg^{2+} + 2e^-$	+2.375	- 2. 37 5
$Al \rightleftharpoons Al^{3+} + 3e^-$	+1.670	- 1. 670
$Mn \rightleftharpoons Mn^{2+} + 2e^-$	+1.029	- 1. 02 9
$Zn \rightleftharpoons Zn^{2+} + 2e^-$	+0.762	- 0. 762
$Cr \rightleftharpoons Cr^{3+} + 3e^-$	+0.740	- 0. 740
$Cr \rightleftharpoons Cr^{2+} + 2e^-$	+0.557	- 0. 557
$Cr^{2+} \rightleftharpoons Cr^{3+} + e^-$	+0.410	-0. 410
$Fe \rightleftharpoons Fe^{2+} + 2e^-$	+0.409	- 0.409
$Cd \rightleftharpoons Cd^{2+} + 2e^-$	+0.402	-0.402
$Co \rightleftharpoons Co^{2+} + 2e^-$	+0.280	-0.280
$Ni \rightleftharpoons Ni^{2+} + 2e^-$	+0.230	-0.230
$Pb \rightleftharpoons Pb^{2+} + 2e^-$	+0.126	-0.126
$H_2 \rightleftharpoons 2H^+ + 2e^-$	Zero	Zero

$Sn^{2+} \rightleftharpoons Sn^{4+} + 2e^-$	-0.150	+0.150
$Cu \rightleftharpoons Cu^{2+} + 2e^-$	-0.340	+0.340
$4OH^- \rightleftharpoons 2H_2O + O_2 + 4e^-$	-0.401	+0.401
$Ag \rightleftharpoons Ag^+ + e^-$	-0.800	+0.800
$Pt \rightleftharpoons Pt^{2+} + 2e^-$	-1.200	+1.200
$Au \rightleftharpoons Au^{3+} + 3e^-$	-1.420	+1.420
$2F^- \rightleftharpoons F_2 + 2e^-$	-2.870	+2.870

Table (1) [For illustration only]

GALVANIC CELLS AND PRODUCTION OF ELECTRIC ENERGY

The galvanic cells are divided according to their nature to produce electrical energy to:

1- **Primary cells.** **2- Secondary cells**.

All of them are systems produce electrical energy through the spontaneous oxidation – reduction reaction.

1- Primary cells

These are saving energy systems in the form of chemical energy which can be converted to electrical energy through spontaneous irreversible oxidation reduction reaction. This kind of cells stops to work when the anode material is consumed and the ions of the cathode half- cell are diminished.

Primary cells not only easy but also impossible to be recharged to returning its constituents to its initial state this means that it is an irreversible cells. Of course for the easy use particularly in the moving equipment, it should be in dry form not in liquid. Thus it is known as dry cells, and the cell in the dry form realize a constant

potential for long time working in addition to the possibility of its manufacture in small size .We shall show as follows some of the important kinds of primary cells :

A- Mercury cell:

This cell is manufactured in shape of cylinder or disk – It is characterized by its small volume . Thus it is commonly used in ear phones , clocks and cameras – the positive pole is zinc and the negative pole is from mercury oxide – potassium hydroxide is used as an electrolyte and the cell is well closed , fig .(4-4) and the total reaction occuring in this cell .

Oxidation

$$Zn^0 + HgO \rightarrow ZnO + Hg^0$$

Reduction

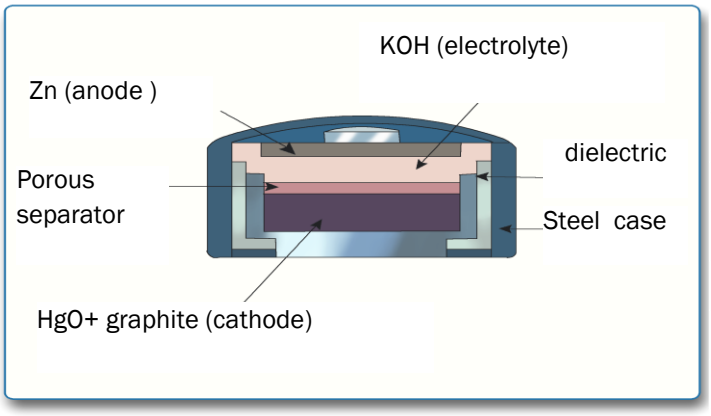


Fig4-4

This cell gives electromotive force E_{cell} = 1.35 volt ,
and it should get rid of this battery after using by a safe way since it contains mercury which is poison material.

2- Secondary cells

These galvanic cells are characterized by reversible chemical reactions and store the electrical energy as chemical energy which can be converted once time to electrical energy when we needed and can be recharged by passing an electric current from external source between their poles in a direction opposite to the discharge process. There are two examples of secondary cells.

A-Lead – Acid Battery :

This kind of batteries are developed and become the suitable kind of batteries used in cars (so is known as car battery) .It consists of six cells are connected in series. Each cell produces **E_{cell} = 2 volt** and the total cell potential of the battery is **emf = 2x6=12**

volts and it is possible to manufacture large volume batteries contain more than six batteries when it is needed.

In this battery the **anode** is network of lead filled with spongy lead (Pb) (fig 4-6) and the cathode is lead network filled with a paste of lead dioxide (PbO_2) and the plates are separated away a way from each other by insulting sheets and all of which are dipped in dilute sulphuric acid solution as a conducting electrolyte all of which are put in a container made of solid rubber or from plastic materials (polystyrene) which does not affect by acids and the battery works as galvanic cell during its operation (discharge) and consuming its energy , while in the case of recharging it is considered as an electrolytic cell . fig (4-6).

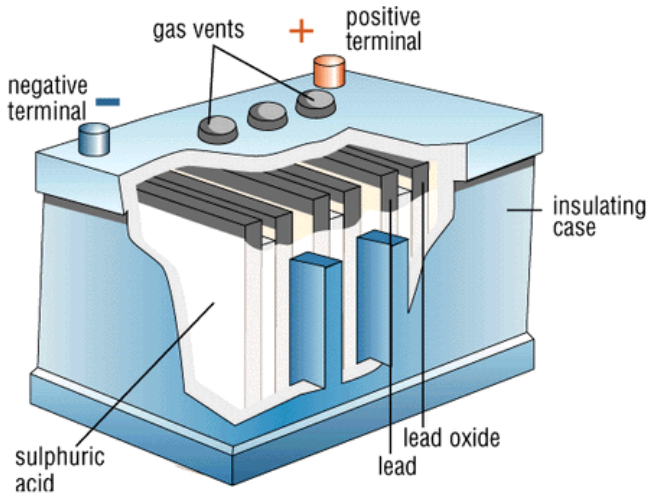


Fig . (4-6) the lead acid battery (lead accumulator

A- Discharging reaction

The total battery reaction occurs during battery discharge :

$$Pb_{(s)} + PbO_{2\,(s)} + 4H^{+}_{(aq)} + 2SO^{2-}_{4\,(aq)} \xrightarrow{\text{discharging}} 2PbSO_{4\,(s)} + 2H_2O_{(l)} \qquad E_{(all)}=2.05V$$

The battery condition can be identified by measuring the density of the acid solution using the **hydrometer** (measuring liquids density) when the battery is completely charged. **The density of acid** equal from 1.28 to1.30 gm/cm^3 and if the acid density decreased to lower than 1.2 gm/cm^{-3} , this means that the battery needs to recharging and increasing its acid concentration .

B-Charging reaction

The long period for using the battery leads to decrease the concentration of sulphuric acid as a result of increasing the quantity of water produced from the reaction and the conversion of cathode material (PbO_2) and anode (Pb) to lead (II) sulphate which lead to decrease of the quantity of electric current produced from it . Here the battery needs to recharging and this is done by connecting the battery poles to an outside source of direct electric current

Whose potential is slightly higher than the potential produced from the battery this will lead to the reversing of the spontaneous reaction which occurred during the discharge of the battery and lead to conversion of lead sulphate to lead at the anode and lead dioxide at the cathode and the concentration of the acid renewed as it was. .

$$\underset{(s)}{2PbSO_4} + \underset{(l)}{2H_2O} \xrightarrow{\text{charging}} \underset{\text{(anode)}}{Pb_{(s)}} + \underset{\text{(cathode)}}{PbO_{2(s)}} + \underset{(aq)}{2SO_4^{2-}} + \underset{(aq)}{4H^+}$$

The battery is working during charging as an electrolytic cell, where a nonspontaneous chemical reaction has occurred by passing electric current . This means the storage of electric energy which comes from the external source in the form of chemical energy. This is why the secondary cells are considered as storage batteries and in the car the dynamo in a continuous way is used in recharging the battery first by first.

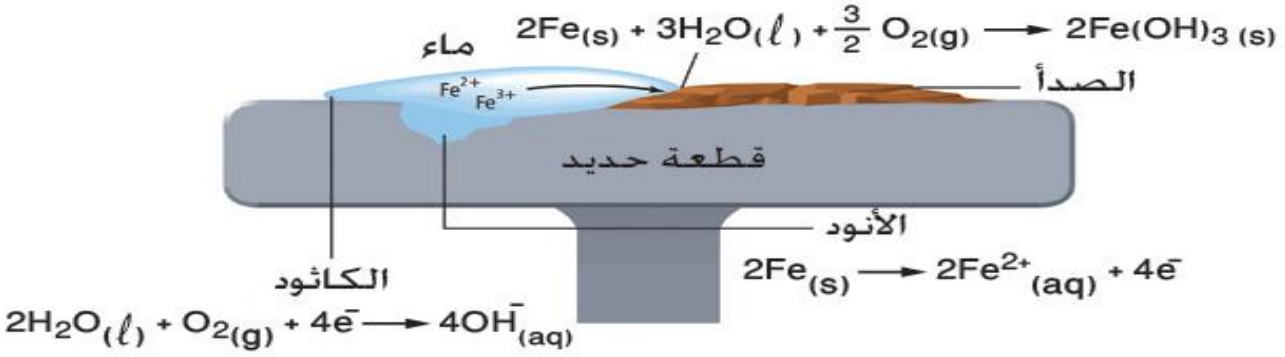
Lesson Three: Corrosion of metals

The corrosion of metals

Cause metal corrosion in heavy economic losses lead to deterioration of metal structures specially that are made from iron , the mass of iron that may be lost due to the corrosion can be estimated by one quarter of the production of the whole world every year . Hence the interest in this phenomenon and try to overcome and any chemical erosion of metals in known as the rust by environment effects.

The mechanism of corrosion:

In most cases the corrosion of the pure metals is difficult. Iron as example doesn't rust easily in its very pure state but most industrial metals contain different impurities that activate the corrosion process , the connection between less active metal to the more active metal causes corrosion for the more active in the same medium, then we can conclude that the corrosion of metals is formed due to the formation of galvanic cells in which the anode is the corroded element while the cathode is the is the less active one or carbon that exists as impurities which causes corrosion.



(Fig 4-8)

When a piece of steel is exposed to cracking or breaking it will form a galvanic cell with the water which contains dissolved ions that acts as electrolytic solution and iron acts as anode.

We obtain the total equation for cell of iron corrosion.

$$2Fe_{(s)} + 3H_2O_{(\ell)} + 3/2\ O_{2(g)} \longrightarrow 2Fe(OH)_{3(s)}$$

The rust is a slow process due to the presence of small amount of ions in water and it increases if a big amount of ions exists in water as in sea water.

The factors that causes corrosion:

These factors can be classified into two kinds:

1- Factors specific to the metal. 2-Factors specific to the surrounding.

1- Factors specific to the metal:

a- Heterogeneity of alloys: the metals that used in industry may be exist as alloys which are difficult to be prepared homogenous structure , so a large number of local cells are formed which cause corrosion of the mare active metal.

b- The connection of metals: at positions of welding metals together or using screws made from another metal, this causes the formation of local galvanic cells causes corrosion of active metal , as the connection of aluminum and copper, aluminum is corroded firstly and the connection of iron with copper iron is corroded firstly.

2-The external factors (Factors specific to the surrounding):

Water, oxygen and salts are considered from the external factors that causes metal corrosion.

Protection of metal from corrosion:

The protection and preventing metals from corrosion specially iron, is considered from the bases of the international economy.

Some methods the protection of iron from rusting, by covering it by another substance to to be away for the surrounding this is formed by one of two methods:

1- Painting iron by the organic material like oil or varnish or slakon but it is not effective method on the long run.

2- Covering iron by metals resist corrosion as galvanizing (iron galvanizing)

By sinking steel in molten zinc, also magnesium is used to protect iron that used in ships industry , tin is used ito protect iron that used in cans

a- Cathode protection (cathode cover):

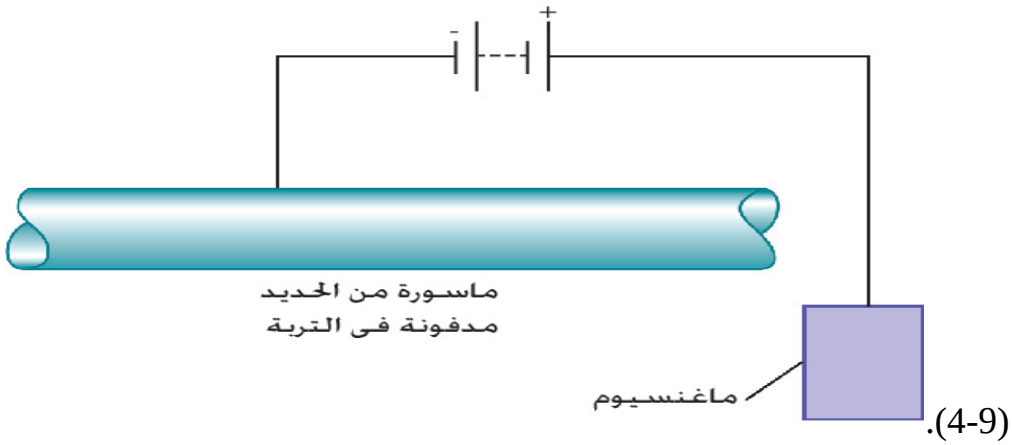
If the protective element less active (like ten) than the original element (iron) a galvanic cell will be formed in which iron becomes the active metal and anode while tin becomes the less active metal and cathode Scratching iron corroded.

-Therefore, iron that painted with tin corroded on scratching faster than iron.

b-Anodic protection: (Anodic cover)

covering the metal by other more active metal such as plating iron by zinc (iron galvanization) where Zn precedes iron in the electromotive series ,then galvanic cell is formed in which Zn acts as anode so it is corroded completely first before iron before starting of iron corrosion which takes a long time as iron corrosion begins at its surface, as ships

chassis are continually in contact with salty water and also iron pipes are buried in moist soil so they are more exposed to corrosion , and can be protected by connecting them to the negative pole of electric sources so they act as cathode and also connect a more active metal than iron like Mg which acts as anode, in this case Mg is corroded instead of iron and called scarifying electrode.



.(4-9)

Lesson Four: The Electrolytic cell

Secondly : Electrolytic cells.

Electrolytic cells: are electric cells in which the electric energy from an external source is converted to chemical energy used through a nonspontaneous oxidation – reduction reaction. As shown in fig .(4-10) .It is a container contains electrolyte solution (like solution to any salt base , acid or melt to any salts) in which two electrodes of the same material (like carbon , or platinum) or each one of them is of different material (like carbon, platinum, copper, zinc or others) One of the electrodes is connected by the positive pole of the battery to become positively charged electrode at which oxidation reactions occur anode , while the other electrode is connected by the negative pole of the battery and at which reduction reactions occur at (cathode) .It is interesting to note that electrolytes which used as conductors in these cells are differ than electronic conductors (metals) . There are two kinds of liquid electrolytes .The first solutions of acids , bases or salts and the second is the molten of salts.

When the electrodes are connected with the applied potential on the cell slightly overcome the cell reversible potential an electric current in the electrolytic cell flows and here the positive ions present in the conducting electrolyte solution between the two electrodes are directed towards the negative electrode (cathode) and neutralize their charge by accepting electrons (reduction) , while the negative ions from the electrolyte solution are directed towards the positive electrode (anode) and neutralize their charge by losing electrons (oxidation). If the electrolytic solution in this cell is copper chloride ($CuCl_2$) , the following oxidation – reduction reactions occur in the electrolytic cell :

a- **Oxidation reaction at the anode** (positive electrode)

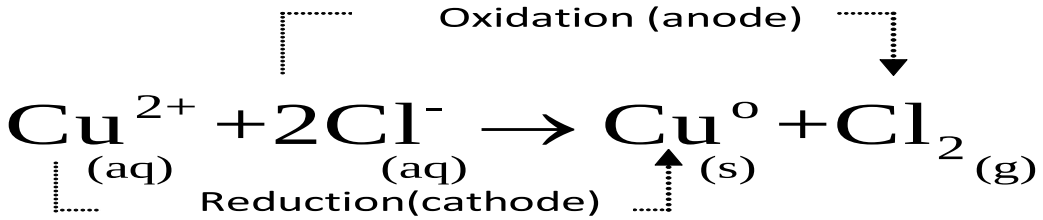
$$2Cl^-_{(aq)} \xrightarrow{\text{oxidation}} Cl_{2(g)} + 2e^- \text{ (oxidation potential= -1.36 volts)}$$

b –

Reduction reaction at the cathode (negative electrode)

$$Cu^{2+}_{(aq)} + 2e^- \xrightarrow{\text{reduction}} Cu^{o}_{(s)} \qquad \text{(reduction potential= +0.34 volts)}$$

and the total reaction occurs in the cell is the sum of the anode and cathode reactions .



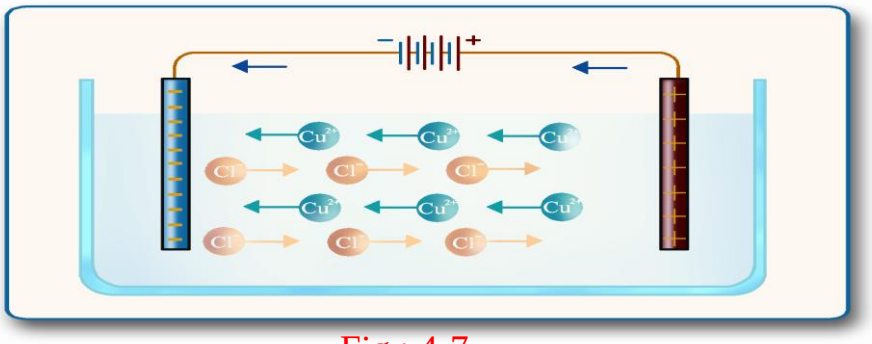


Fig : 4-7

The electrolysis: It is a chemical decomposition of the substance due to the effect of electric current that passes in the electrolyte

Lesson Fife: The Electrolytic cell

APPLICATIONS ON ELECTROLYSIS

1- Electroplating

It is a process of formation of thin layer of certain metal on the surface of another metal to give it a nice shiny view or to protect it from corrosion. We find as example that some cars parts which are manufactured from steel are electroplated by a chromium layer to take a beautiful shape and also to protect it from corrosion. Also some healthy tools like. Water taps, mixers are electroplated by chromium or gold. Also electroplating is used in raising the value of some cheap metals after plating by chromium or gold or silver.

If we take for example, the electroplating of a jug by a layer of silver, the surface should be firstly completely cleaned, then dipped in electrolyte solution contains silver ions (like silver nitrate) and a rod or column of silver metal is put in the solution-the Jug should be connected by the negative pole of the battery, thus it becomes a cathode and the silver metal column is connected by the positive pole, thus it becomes anode in a cell as shown in Fig- (4-12)

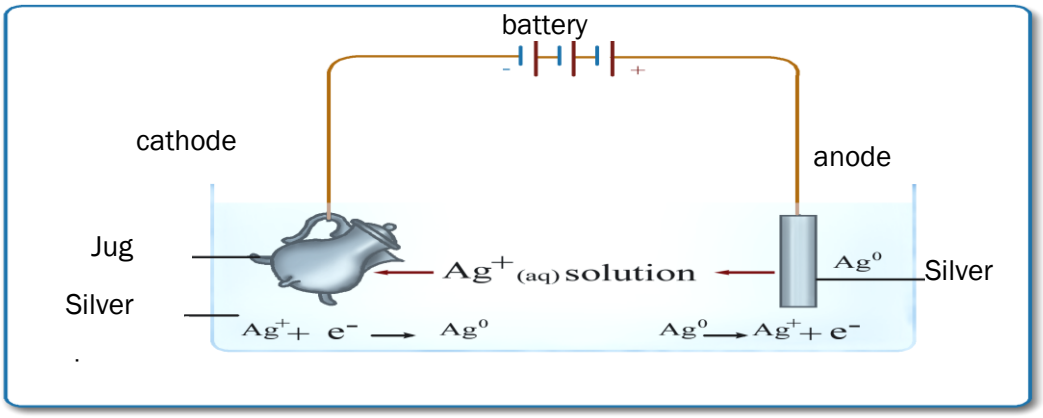


Fig 4: 12-4Electroplating by electricity

2- preparation of aluminum

Aluminum is electrically extracted from bauxite (Al_2O_3) which is dissolved in cryolite melt (Na_3AlF_6) containing a little of fluorspar (CaF_2) to decrease the mixture melting point from 2045^0c to 950 ^{0}c

Recently **cryolite** is exchanged by using a mixture of **fluoride salts of Aluminum, Sodium and Calcium** where this mixture gives with bauxite a melt which is characterizes by a low melting point and a lower density value compared to the melt obtained from cryolite. The lowering of the melt density facilitates the melted aluminum separation which forms a deposited in the bottom of the electrolysis cell.

Fig .(4-13) represents a diagram of the used electrolysis cell.

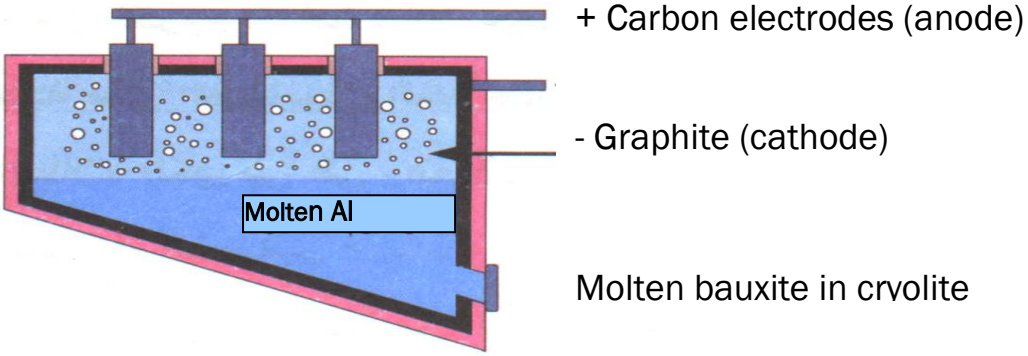


Fig :4-13 preparation of Aluminum by electrolysis

In this cell the cathode is the body of the cell container which is made from iron plated by layer of carbon (graphite) ,while the anode is carbon rods (graphite) . When the electric current passes between the cell electrodes an oxidation – reduction reaction occurs.**and the total reaction is :**

$$2Al_2O_3 \xrightarrow{\text{electrolysis}} 4Al + 3\,O_2$$

Then aluminum is withdrawn from the cell through a special opening.

3- Purification of Metals

The degree of purity of metals which are prepared in industry is lower than the required degree of purity which is required its efficiency.

For example, copper of 99% purity contains iron, zinc and gold impurities which decrease the ability of copper to conduct electrical and also its quality. Thus electrolysis method for purification is used for copper that used in manufacturing electrical wires. The anode (positive pole) in the electrolytic cell is an impure copper metal (Cu^0) , while the cathode (the negative pole) is from wires or fine copper sheets and the electrolyte is an aqueous copper sulphate molecules which dissociates in water to copper ions (Cu^{2+}) and sulphate (SO_4^{2-}) when electric current passes from the external battery at a potential slightly more than the standard potential of half copper cell , the ions are directed towards electrodes which are opposite in the charge.

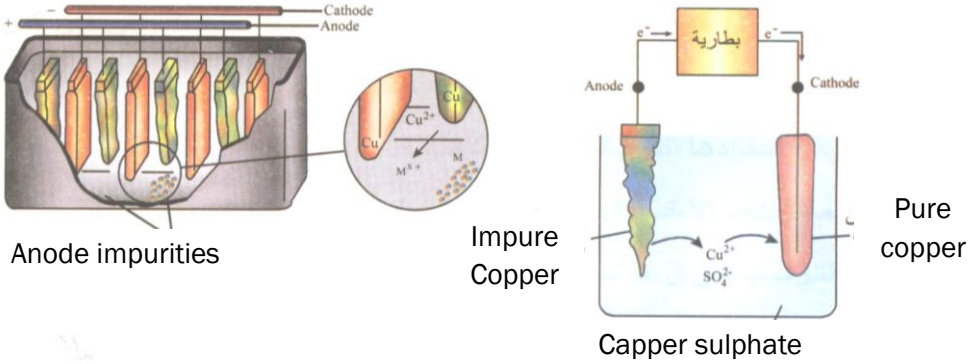


Fig 4-14 Purification of copper electrically

At the anode copper dissolves (oxidizes) and converts to copper ions (Cu^{2+}) which spread in solution and returns to deposit in the form of pure copper at the cathode.

At anode: $Cu_{(s)} \longrightarrow Cu^{2+}_{(aq)} + 2e^-$

At cathode: $Cu^{2+}_{(aq)} + 2e^- \longrightarrow Cu_{(s)}$

While the impurities originally present in the anode material, some of it dissolves (oxidizes) in the solution like iron and zinc but do not deposit at the cathode because they have low reduction potential if it is compared with copper ions,

$$Zn^{0}_{(s)} \longrightarrow Zn^{2+}_{(aq)} + 2e^{-}$$

$$Fe^{0}_{(s)} \longrightarrow Fe^{2+}_{(aq)} + 2e^{-}$$

While silver and gold impurities if it is present in the anode material it does not oxidize (do not dissolve) at the oxidation potential of copper and failed down the anode and removed from the bottom of the cell and by this method we can obtain 99.95 purity of copper. This is in addition to the possibility of separation of some rich minerals e.g gold and silver from copper ores as shown in F.g . (14-4)

Lesson Six: Evaluation

QUESTION 1:-

Choose the correct answer between the two brakets :

1-Substances that conduct electric current through the movement of its ions are ..... conductors (metallic – electrolytic – electronic)

2-Systems in which the conversation of chemical energy to electrical energy is completed as result of occuring a spontaneous oxidation and reduction is.....(electrolytic cells – galvanic cells – solar cells) .

3-The electrode at which reduction process occurs in galvanic cells is..... (Negative electrode – anode – cathode)

4-The electrode at which oxidation process in electrolytic cells occur is..... (Negative electrode – anode – cathode)

5-The moving material particles in melts or solution which are rich in electrons are (positive ions – negative ions- molecules) .

6-The standard hydrogen electrode has value of ..... (zero – positive – negative) .

QUESTION

If you have a copper spoon what are the different steps you must follow it to plate it by a layer of silver – write down the equations (reactions) which occur at each of the cathode and anode .

QUESTION

How we can obtain Aluminium from bauxite.

QUESTION

Pure copper 99% contains impurities. show how it can be purified from impurities to obtain 99.95 purity degrees.

QUESTION

How can you obtain pure gold from a copper wire contains impurities of gold ?

QUESTION

Compare between each of the cathode and anode in the galvanic and electrolytic cells?